

Physical Layer of IoT Communications

Internet of Things



CS5055 - 2025I

PROF.: CWILLAMS@UTEC.EDU.PE

SRC: LoRaWAN Specification, v1.0.4, 2020.

A. Augustin, J. Yi, T. Clausen, W. M. Townsley. A study of LoRa: Long Range & Low Power Networks for the Internet of Things. *Sensors*, 16(9), 2016.

<https://www.lora-alliance.org/>

H. Betiol, J. F. Borin. Análise de Desempenho de Gateway LoRa/LoRaWAN com dispositivos IoT no Smart Campus, Relatório Técnico, 2019.

Outline

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Introduction

BLE – All layers

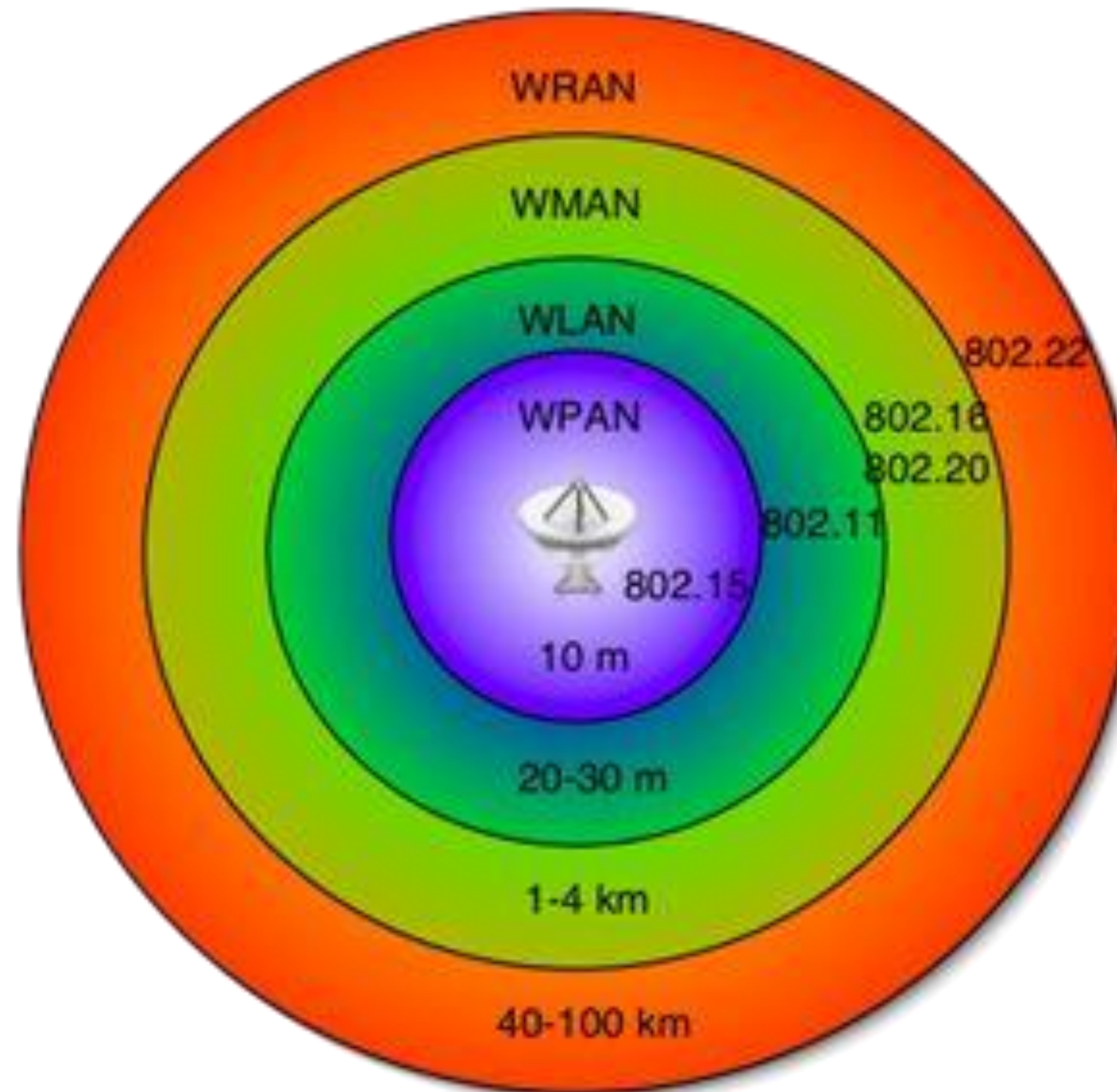
Lora and the Physical Layer Comm

Conclusion

Wireless Personal Area Networks (WPANs)

3

- 10 m or less



Source: <https://www.cse.wustl.edu/~jain/cse574-06/ftp/wimax/index.html>

IEEE 802.15.1 and Bluetooth

- Bluetooth started with Ericsson's Bluetooth Project in 1994 for radio communication between cell phones over short distances.
- Intel, IBM, Nokia, Toshiba, and Ericsson formed Bluetooth Special Interest Group (SIG) in 1998.
- 1st version of the specification came out in 1999.
- IEEE 802.15.1 (2002) is based on Bluetooth specifications.

IEEE 802.15.1 and Bluetooth

- **Bluetooth v1.1** – IEEE 802.15.1-2002
- **Bluetooth v1.2** – IEEE 802.15.1-2005 (Completed Nov. 2003): Faster Connection and Discover, adaptive FHSS – reduce radio frequency interference -, Extended Synchronous Connections – improve voice quality of audio links.
- **Bluetooth v2.0 + Extended Data Rate (EDR) (2004)**: 3Mbps.
- **Bluetooth v2.1 + EDR (2007)**: Secure Simple Pairing to speed up pairing.
- **Bluetooth v3.0 + High Speed (HS) (2009)**: 24 Mbps using Wi-Fi + Bluetooth.
- **Bluetooth v4.0 (2010)**: low energy. Smaller devices requiring longer battery life. New incompatible PHY. Bluetooth Smart.
- **Bluetooth v4.2 (Dec. 2014)**: privacy, data length extension, IP connectivity.
- **Bluetooth 5 (2016)**: 4x range, 2x speed (2Mbps), 8x the broadcasting message capacity, focus on IoT
- **Bluetooth 5.1 (2019)**: direction finding feature, cache, improved advertising channels
- **Bluetooth 5.2 (2019)**: enhanced attribute protocol, LE power control, isochronous channel
- **Bluetooth 5.3 (2021)**: removal of AMP, channel classification enhancement, connection subrating, enhancements to Periodic Advertising, Encryption Key Size Control

Two flavors of Bluetooth

- Bluetooth BR/EDR - establishes a relatively short-range, continuous wireless connection, which makes it ideal for use cases such as streaming audio.
- Bluetooth with low energy functionality (LE) - allows for short bursts of long-range radio connection, making it ideal for Internet of Things (IoT) applications that don't require continuous connection but depend on long battery life.
- Dual-Mode - dual-mode chipsets are available to support single devices such as smartphones or tablets that need to connect to both BR/EDR devices (such as audio headsets) and LE devices (such as wearables or retail beacons).
- **Source:** <https://www.bluetooth.com/specifications/bluetooth-core-specification>

Outline

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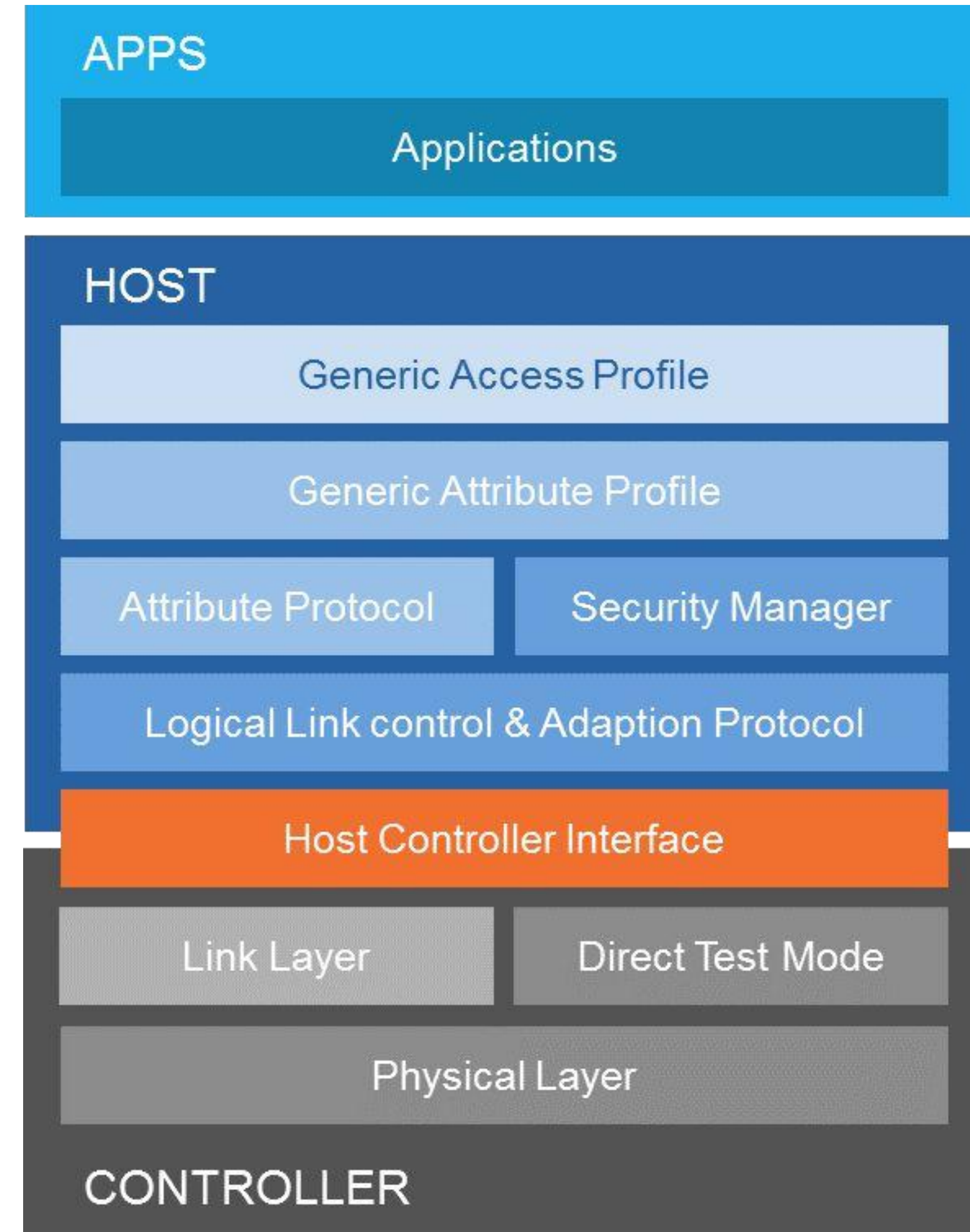
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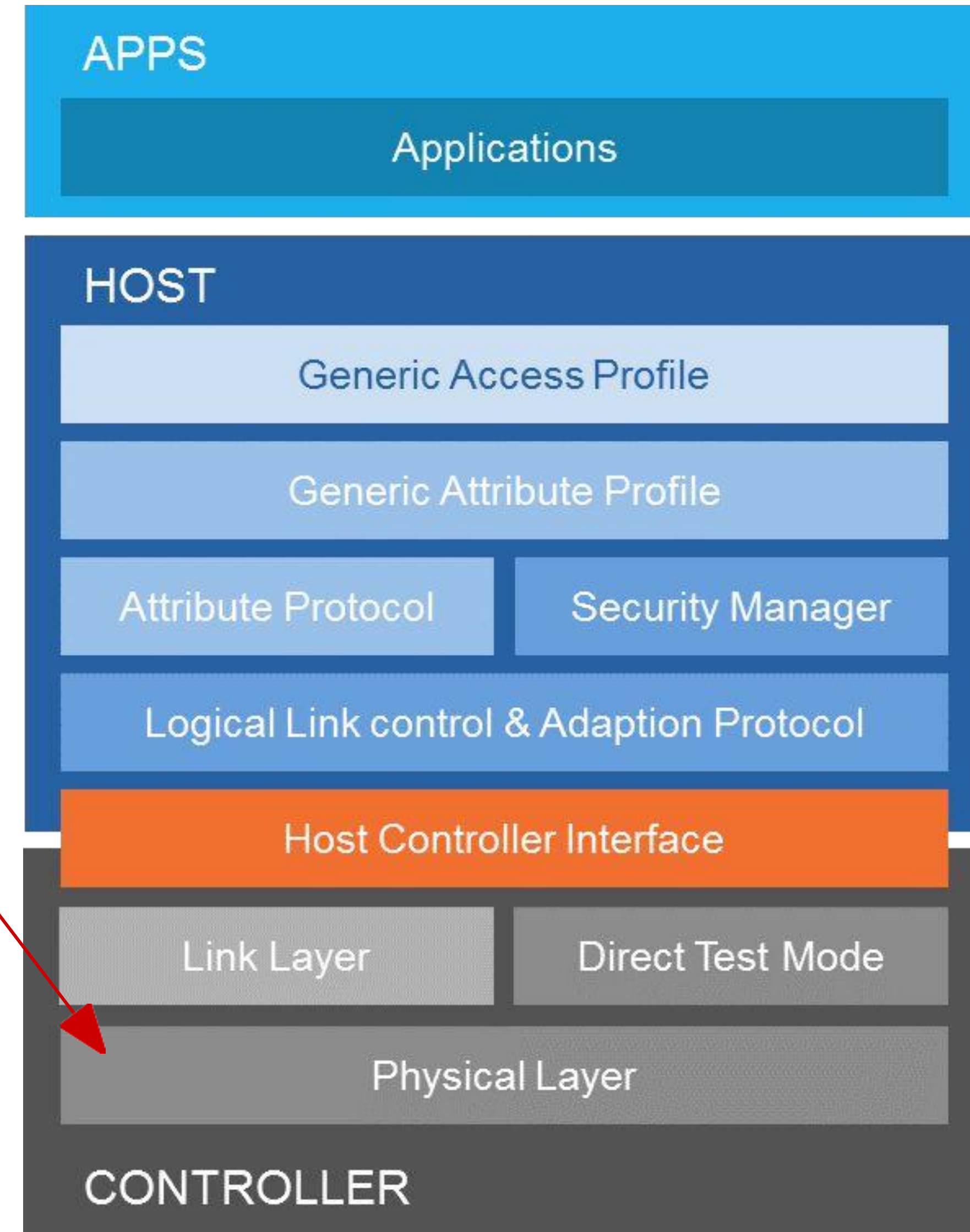
BLE System Architecture



Source: <https://www.bluetooth.com/specifications/bluetooth-core-specification>

BLE System Architecture

Controls transmission/receiving of the 2.4Ghz radio with Bluetooth communication channels. BR/EDR provides channels (79) with narrower bandwidth, while LE uses fewer channels (40) but broader bandwidth.



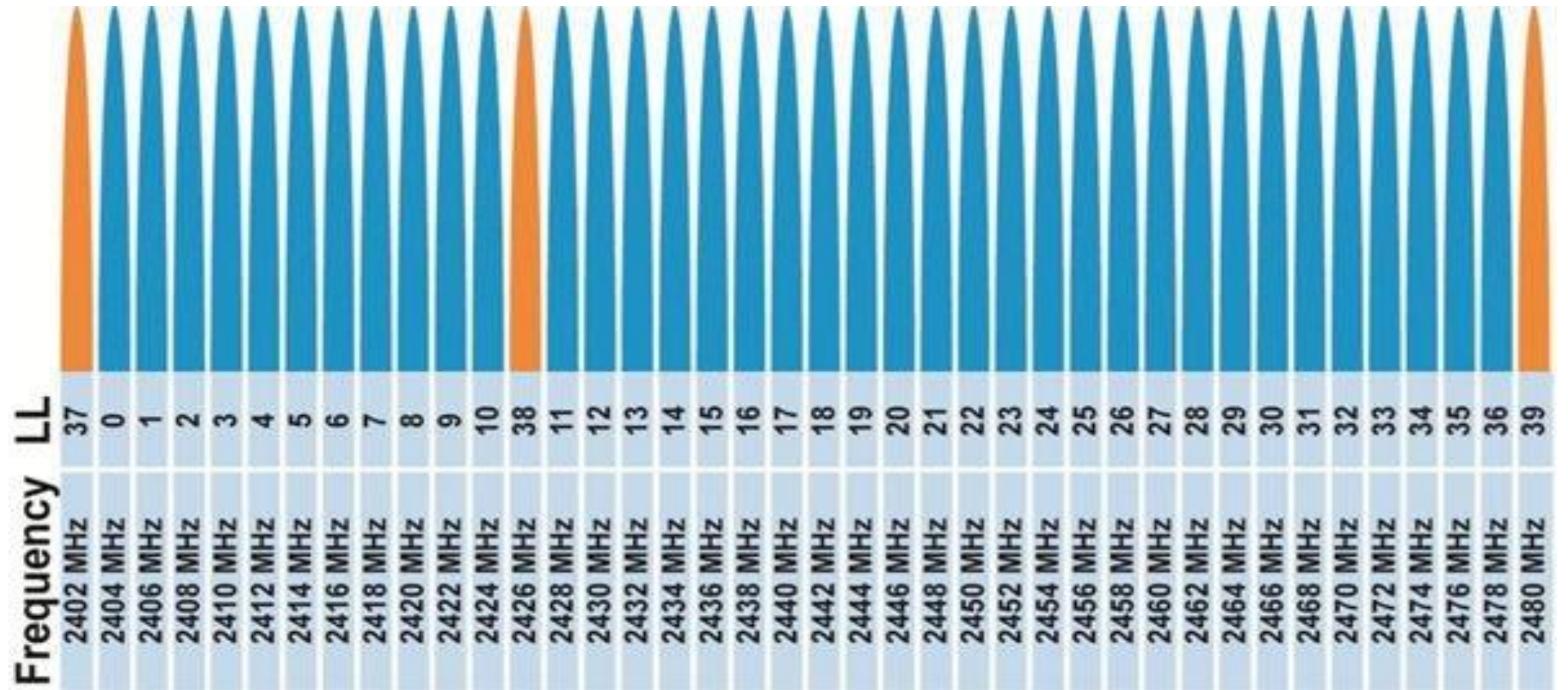
Source: <https://www.bluetooth.com/specifications/bluetooth-core-specification>

Physical Layer: BLE 4.x

10

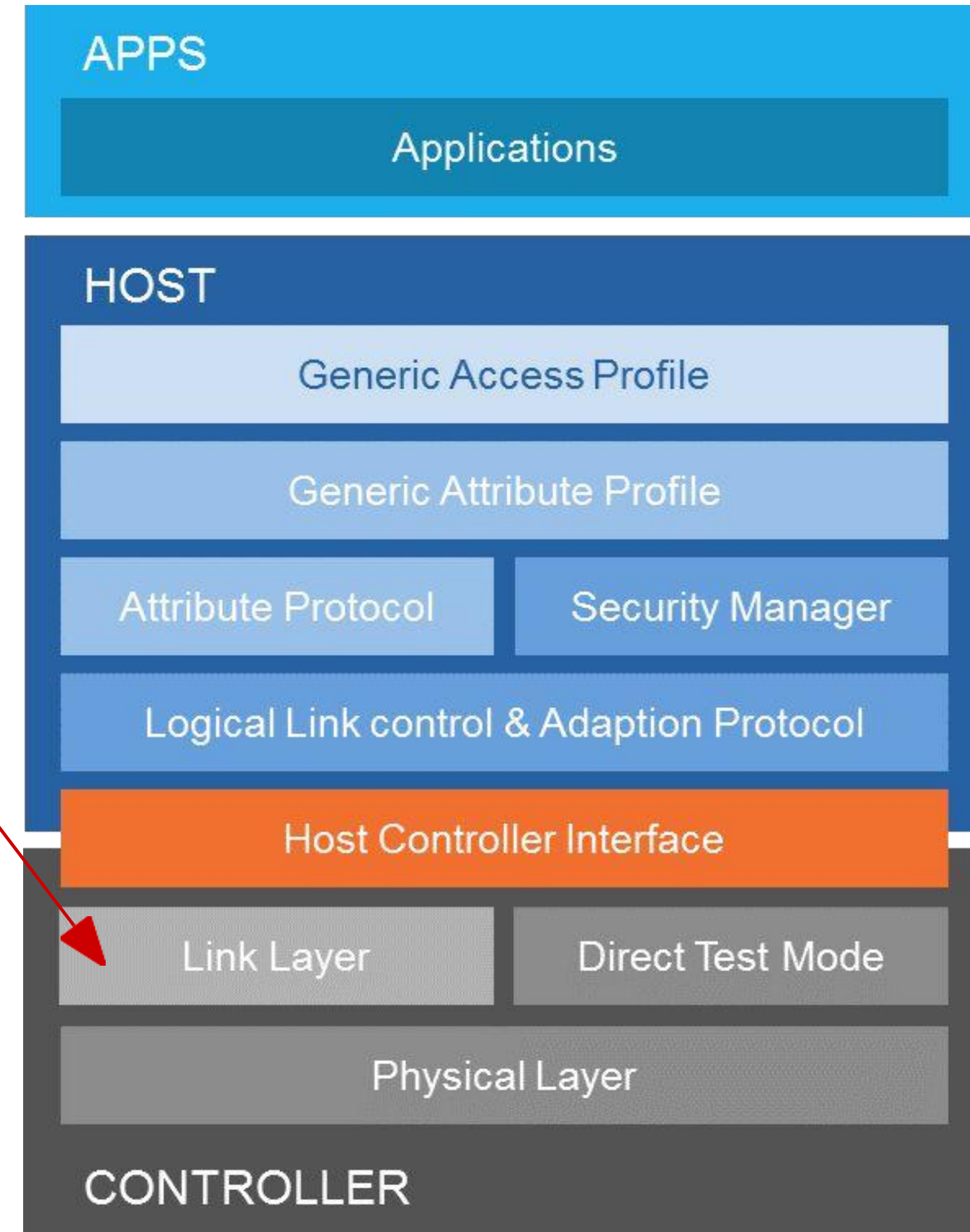
- 2.4 GHz ISM
- 1 Mbps (*theoretical upper limit, 5-10KB/s in a real scenario*)
- 40 Channels on 2 MHz spacing
- Frequency hopping
- Range: can reach 30 m or more *line-of-sight*, but typically closer to 2 to 5 m.

3 advertising channels and 37 data channels



BLE System Architecture

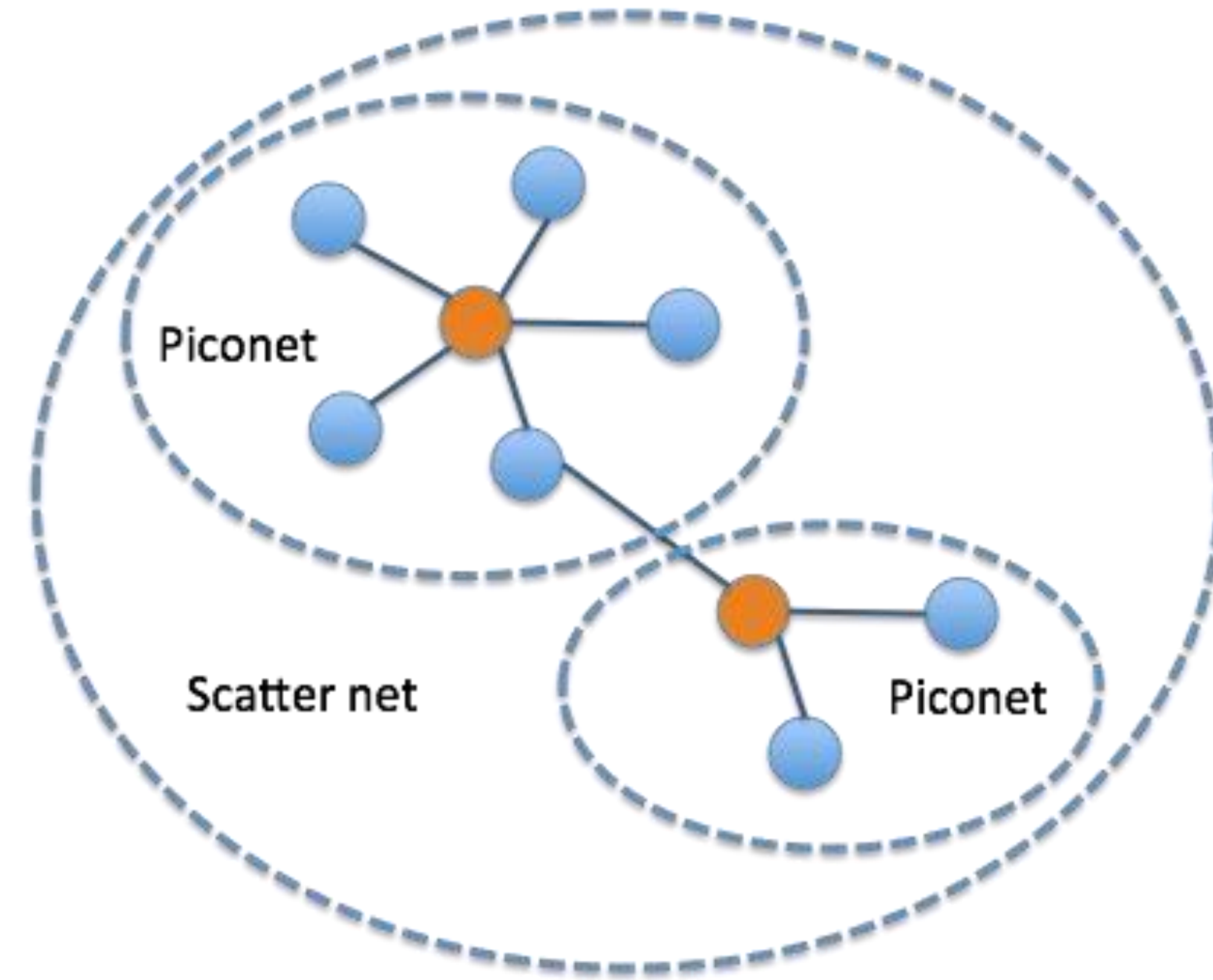
Defines packet structure/channels, discovery/connection procedure and sends/receives data.



Source: <https://www.bluetooth.com/specifications/bluetooth-core-specification>

Bluetooth network

- Piconet:
 - 1 master and many slaves
- Scatter net:
 - A device can participate in multiple Piconets.
 - Routing protocols not defined.



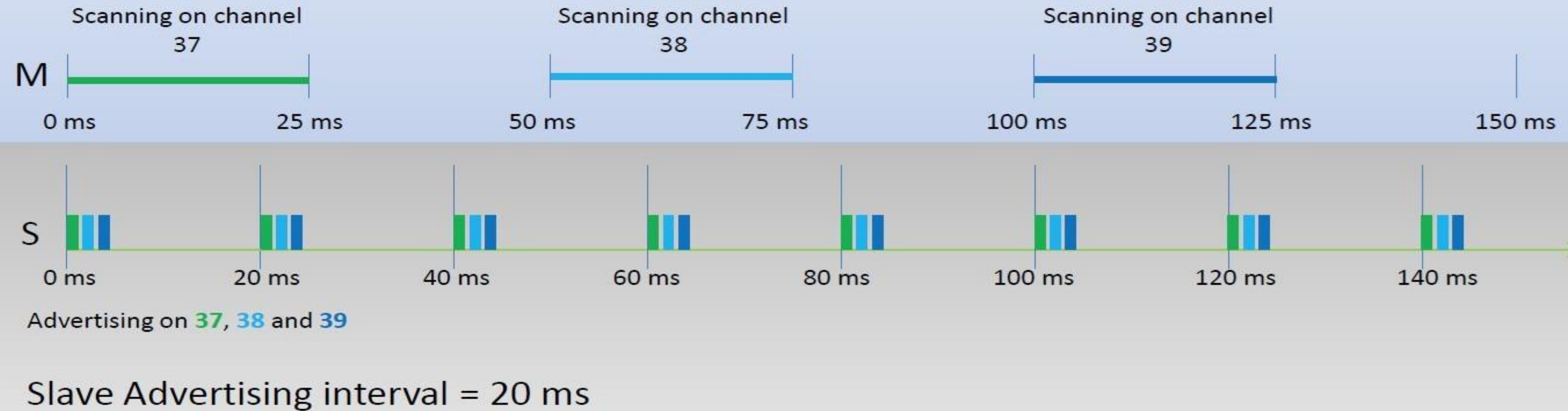
Link Layer: Advertising and Scanning

- One packet format and two types of packets: advertising and data packets
- Advertising packets serve two purposes:
 - To broadcast data for applications that do not need the overhead of a full connection establishment
 - To discover slaves and to connect to them
- Advertising packets are sent at a fixed rate defined by the advertising interval (20 ms to 10.24.s)

Link Layer: Advertising and Scanning

BLE Advertising

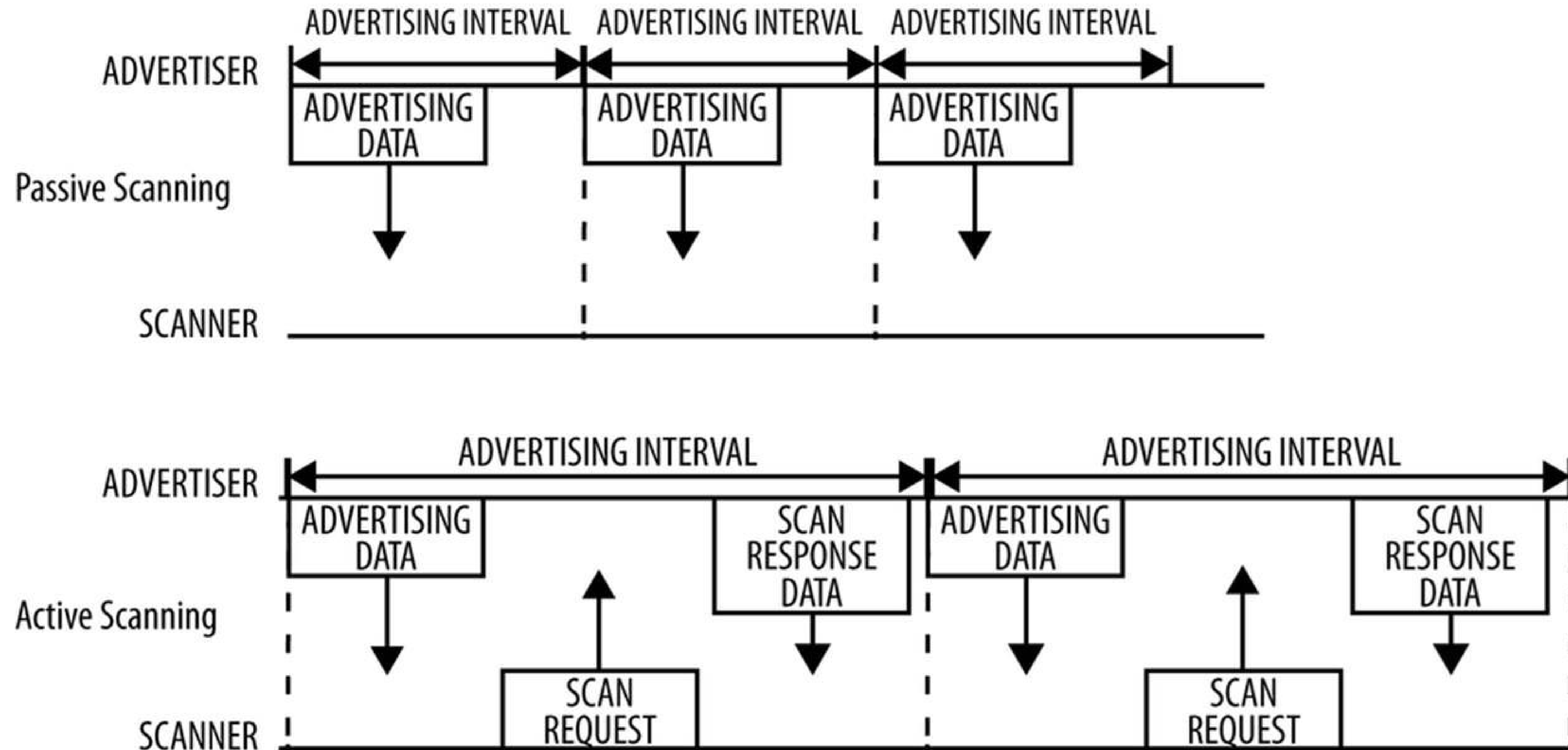
Master scan interval = 50 ms
Master scan window = 25 ms



From this we can see that the following advertising packets will be picked up by the scanning device: $t=0$ $ch=37$, $t=20$ $ch=37$, $t=60$ $ch=38$, $t=100$ $ch=39$ and $t=120$ $ch=39$.

Source: <https://devzone.nordicsemi.com>

Link Layer: Advertising and Scanning



Link Layer: Advertising and Scanning

Classification of advertising packets:

- **Connectability:**
 - Connectable: a scanner can initiate a connection upon reception of such packet
 - Non-connectable: a scanner cannot initiate a connection (broadcast only)
- **Scannability:**
 - Scannable: a scanner can issue a scan request upon reception of such packet
 - Non-scannable: a scanner cannot issue a scan request upon reception of such packet
- **Directability:**
 - Directed: contains only the advertiser's and the target scanner's Bluetooth addresses in its payload. This packet is, therefore, connectable.
 - Undirected: not targeted to any particular scanner.

Link Layer: Connections

Connection establishment:

1. Initiator starts scanning to look for advertisers accepting connections requests.
2. Upon detection of a suitable advertising slave, the initiator sends a connection request packet.
3. If advertiser responds, the connection is established by the initiator. The initiator becomes the master and the advertiser becomes the slave.

Link Layer: Connections

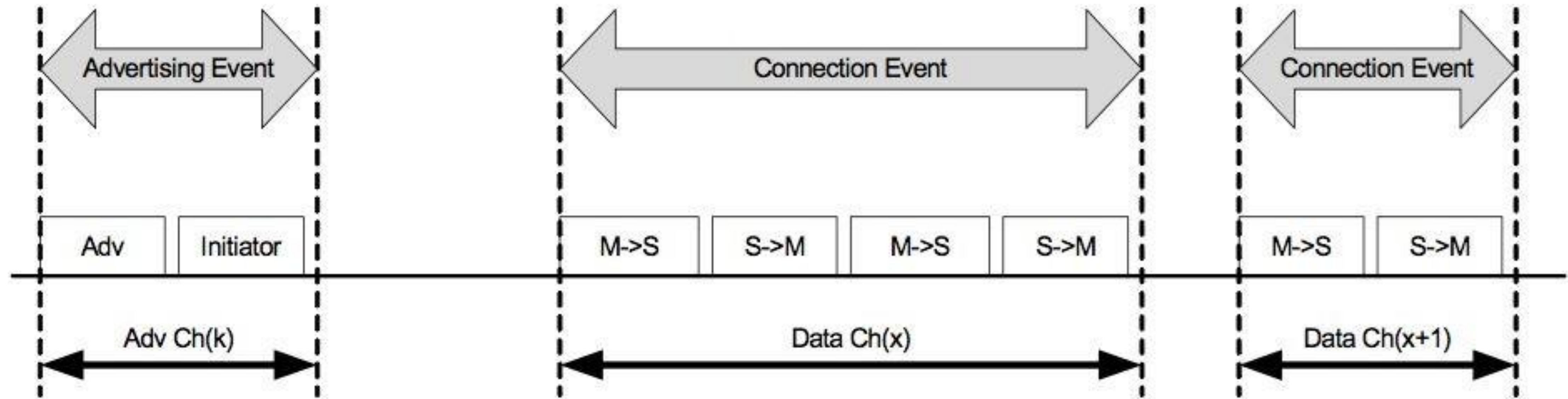


Figure 1.4: Connection Events

Source: Bluetooth Specification Version 4.2.

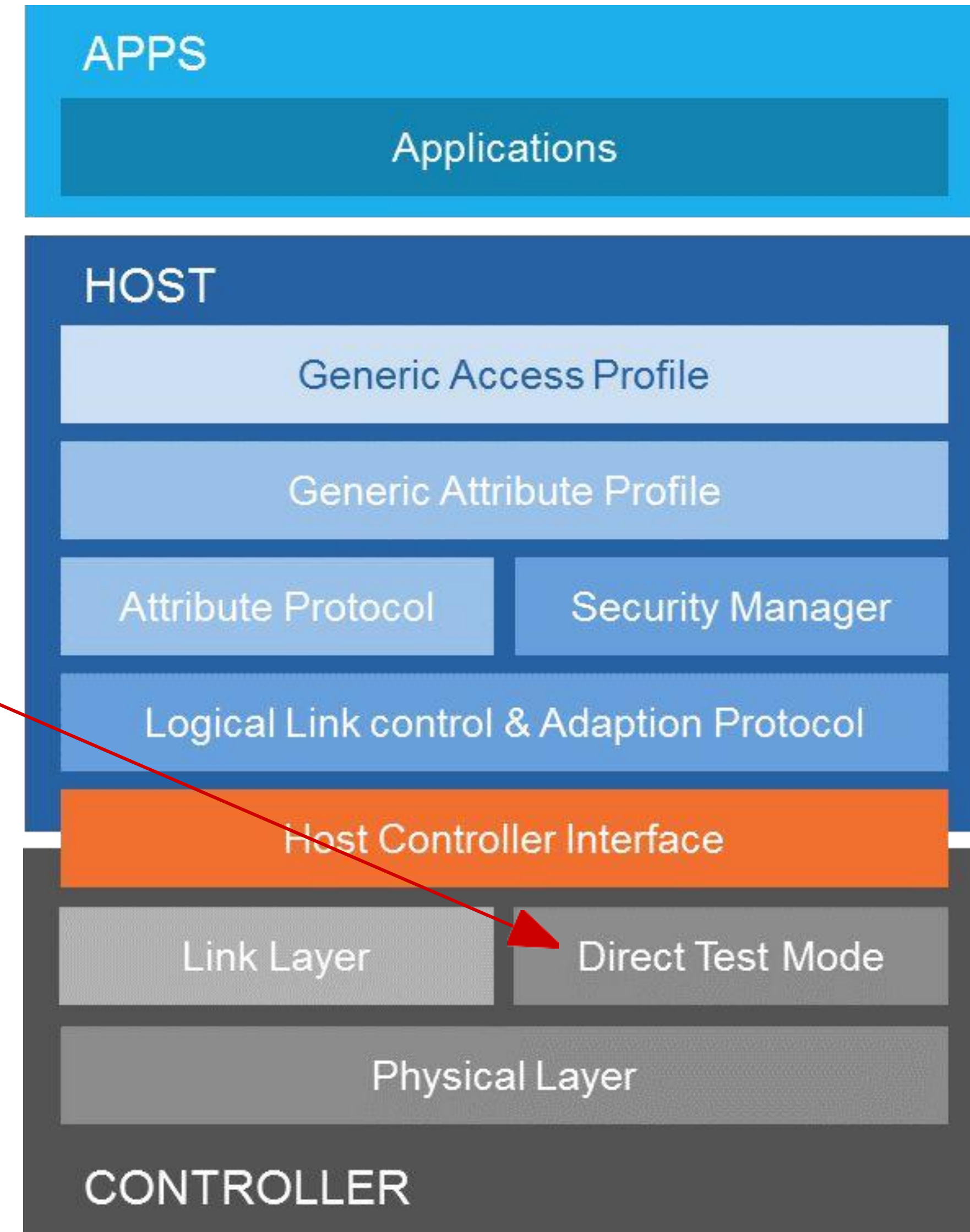
Link Layer: Connections

Connection parameters communicated by the master:

- **Connection interval:** the time between the beginning of two consecutive connection events (7.5 ms to 4s).
- **Slave latency:** number of connection events that a slave can skip without resulting in disconnection.
- **Connection supervision timeout:** maximum time between two received valid data packets before a connection is considered lost.

BLE System Architecture

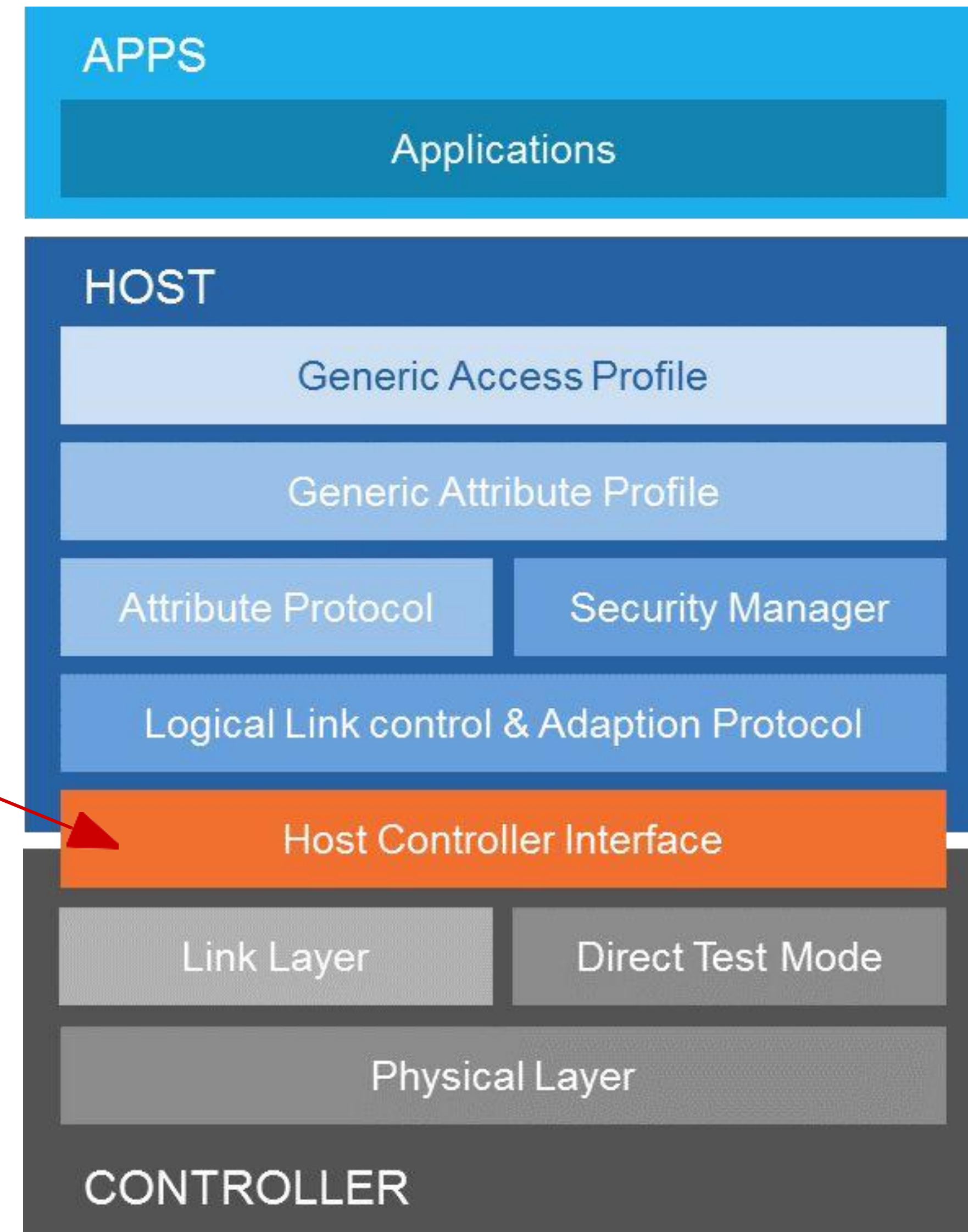
Allows testers to instruct the PHY layer to transmit or receive a given sequence of packets, submitting commands to it either via the HCI or via a 2-wire UART interface.



Source: <https://www.bluetooth.com/specifications/bluetooth-core-specification>

BLE System Architecture

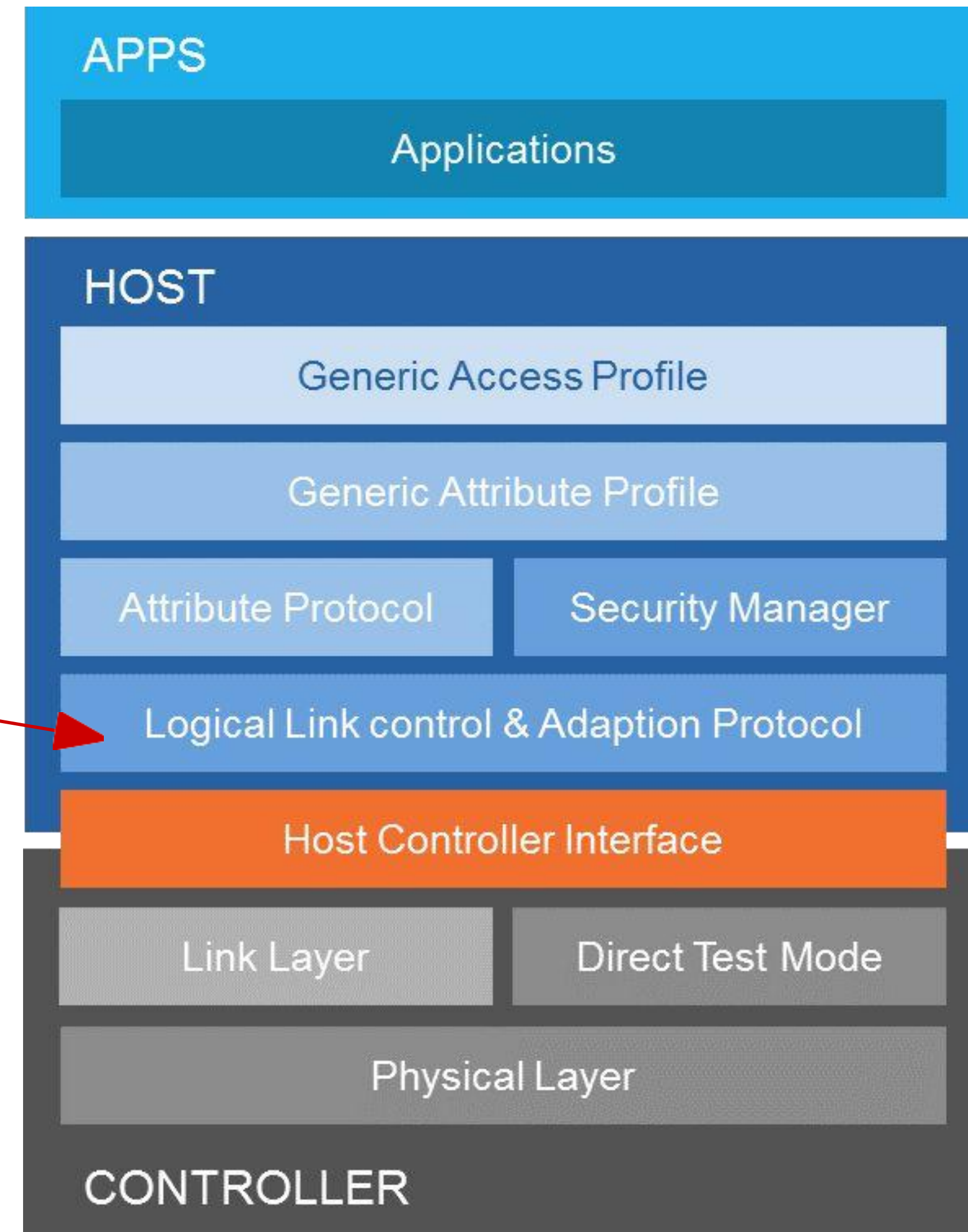
Optional between the standard Bluetooth interface controller subsystem (bottom three layers) and the Bluetooth host.



Source: <https://www.bluetooth.com/specifications/bluetooth-core-specification>

BLE System Architecture

A packet-based protocol that transmits packets to the HCI or directly to the Link Manager in a hostless system. Supports **higher-level protocol multiplexing**, packet segmentation and reassembly, and the conveying of quality of service information to higher layers.

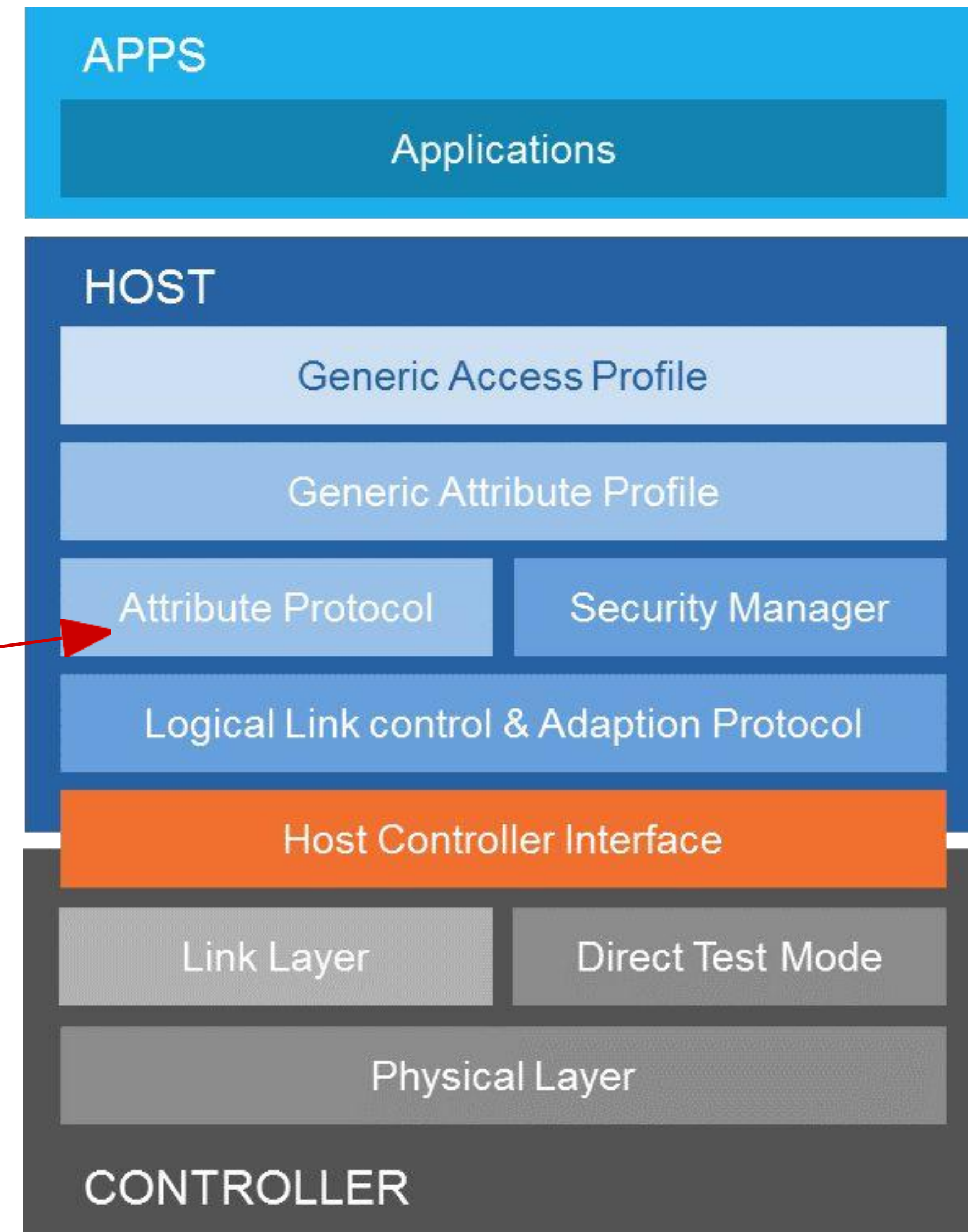


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BLE System Architecture

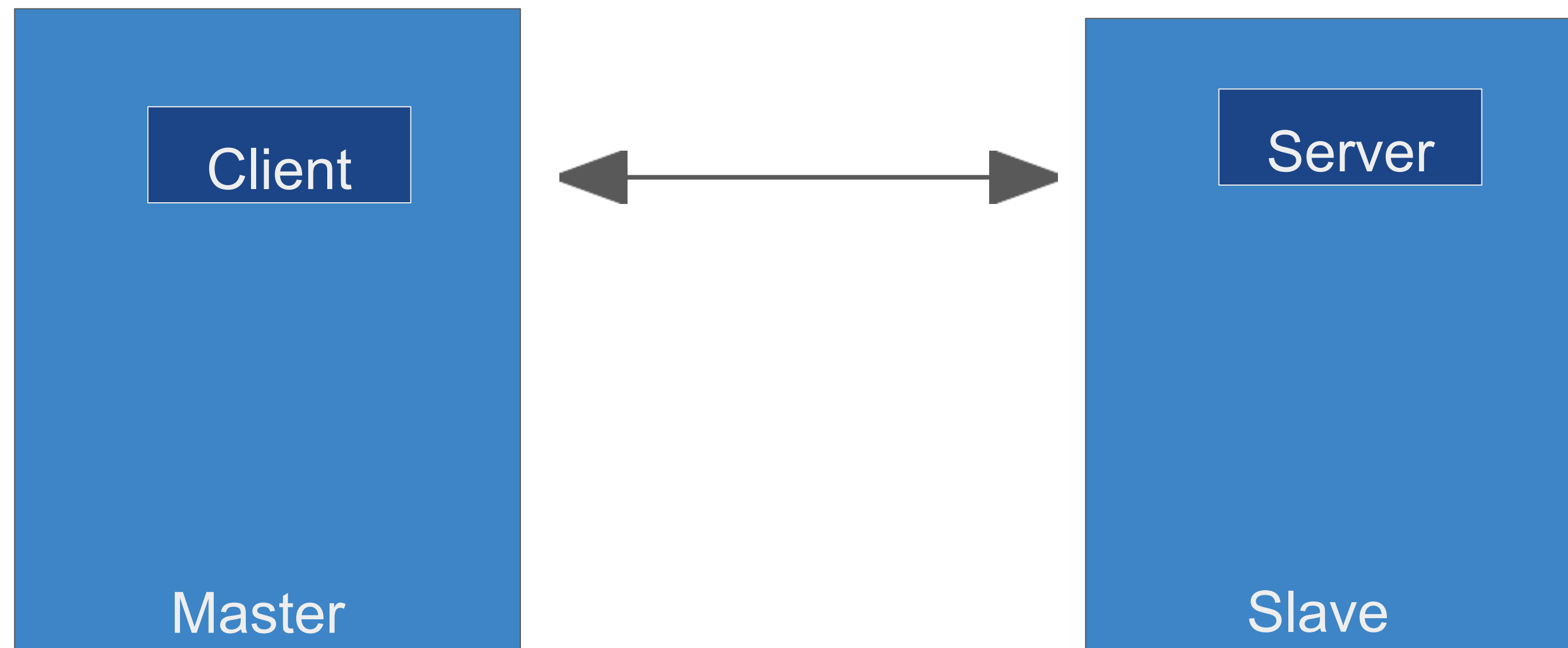
Defines the **client/server protocol** for data exchange once a connection is established.

Attributes are grouped together into meaningful services using the Generic Attribute Profile (GATT). ATT is used in LE implementations and occasionally in BR/EDR implementations.



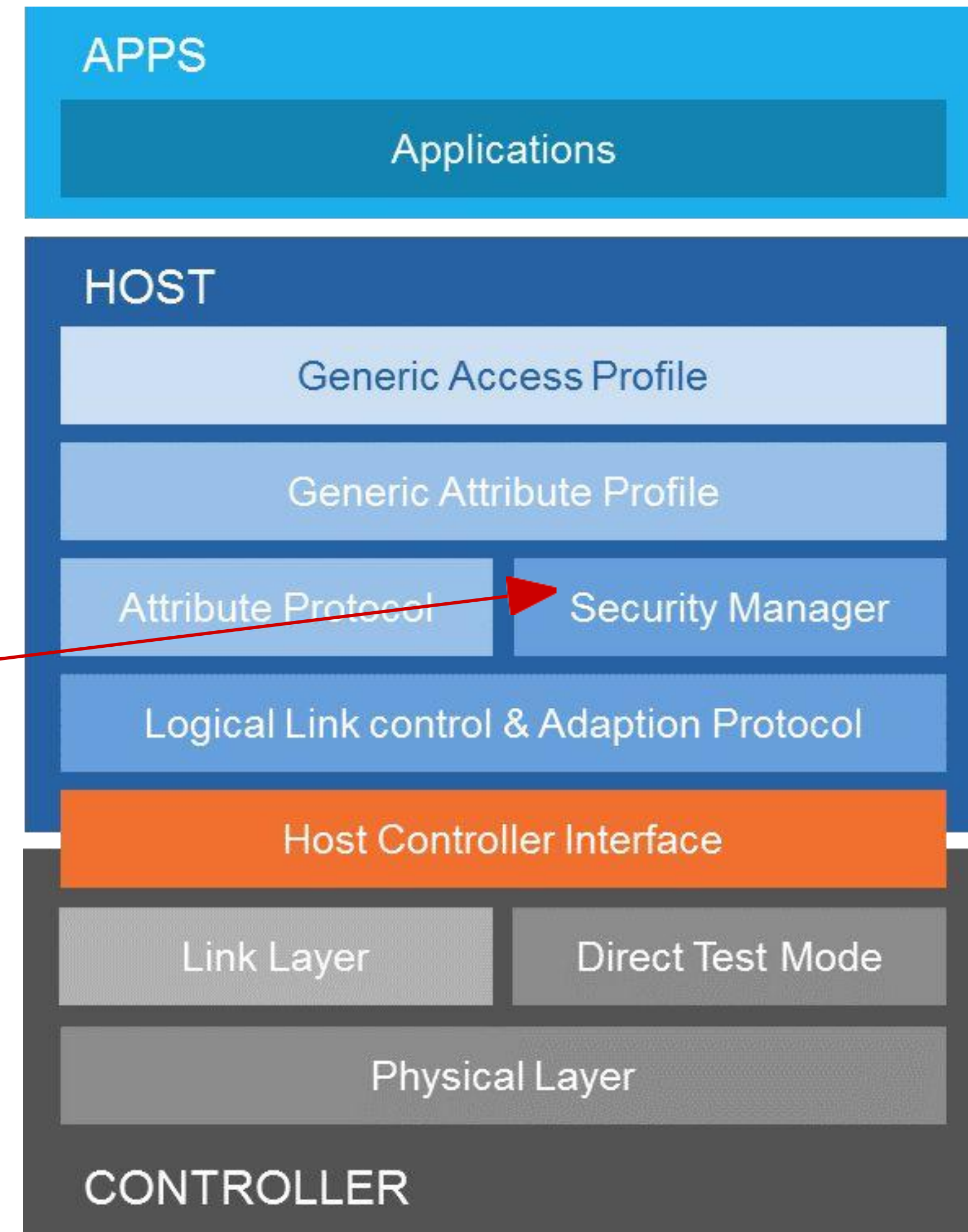
Source: <https://www.bluetooth.com/specifications/bluetooth-core-specification>

Attribute Protocol (ATT): Client-Server



BLE System Architecture

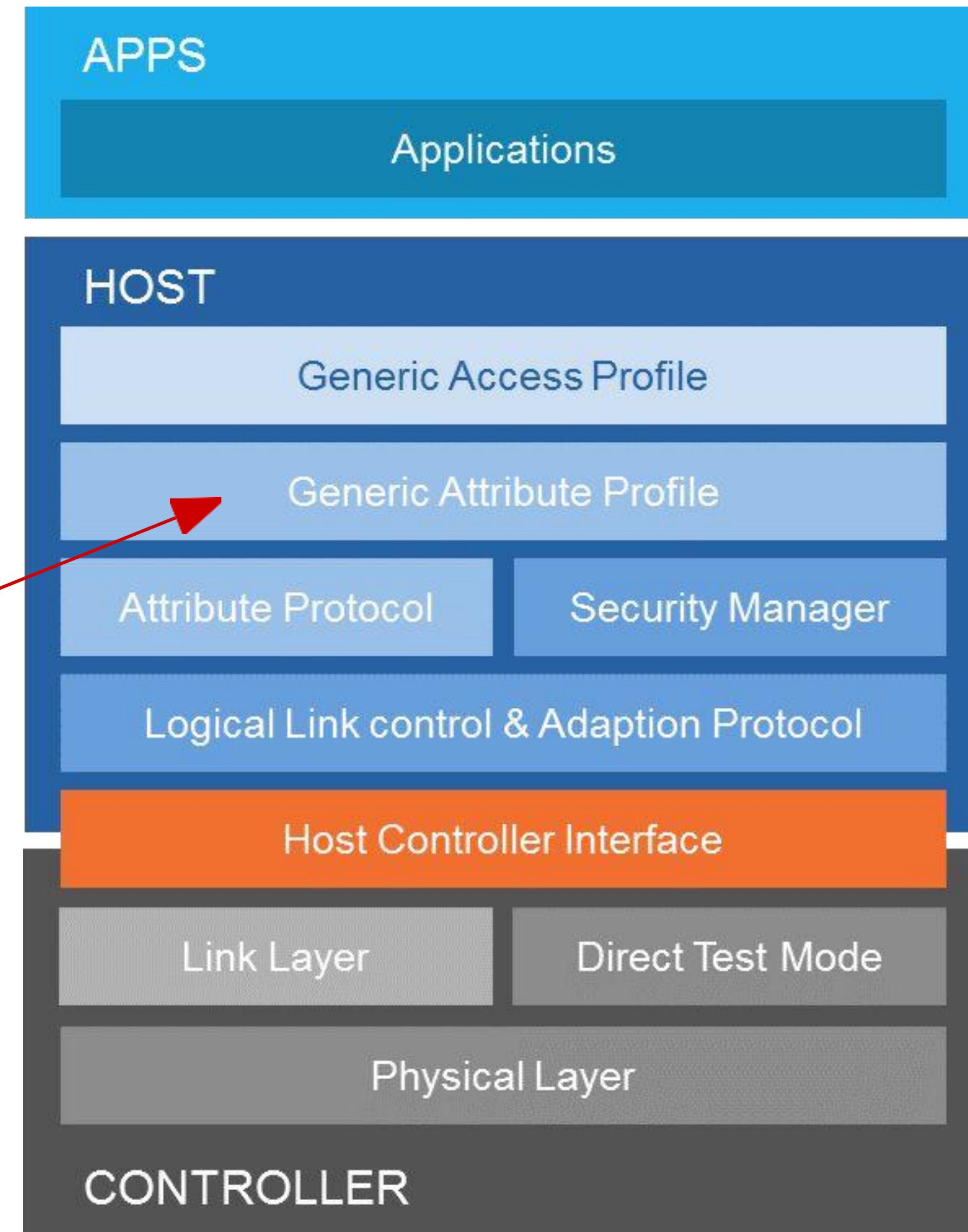
Defines the protocol and behavior that manages pairing integrity, authentication and encryption between Bluetooth devices, and provides a toolbox of security functions that other components use to support almost any level of security needed by diverse applications.



Source: <https://www.bluetooth.com/specifications/bluetooth-core-specification>

BLE System Architecture

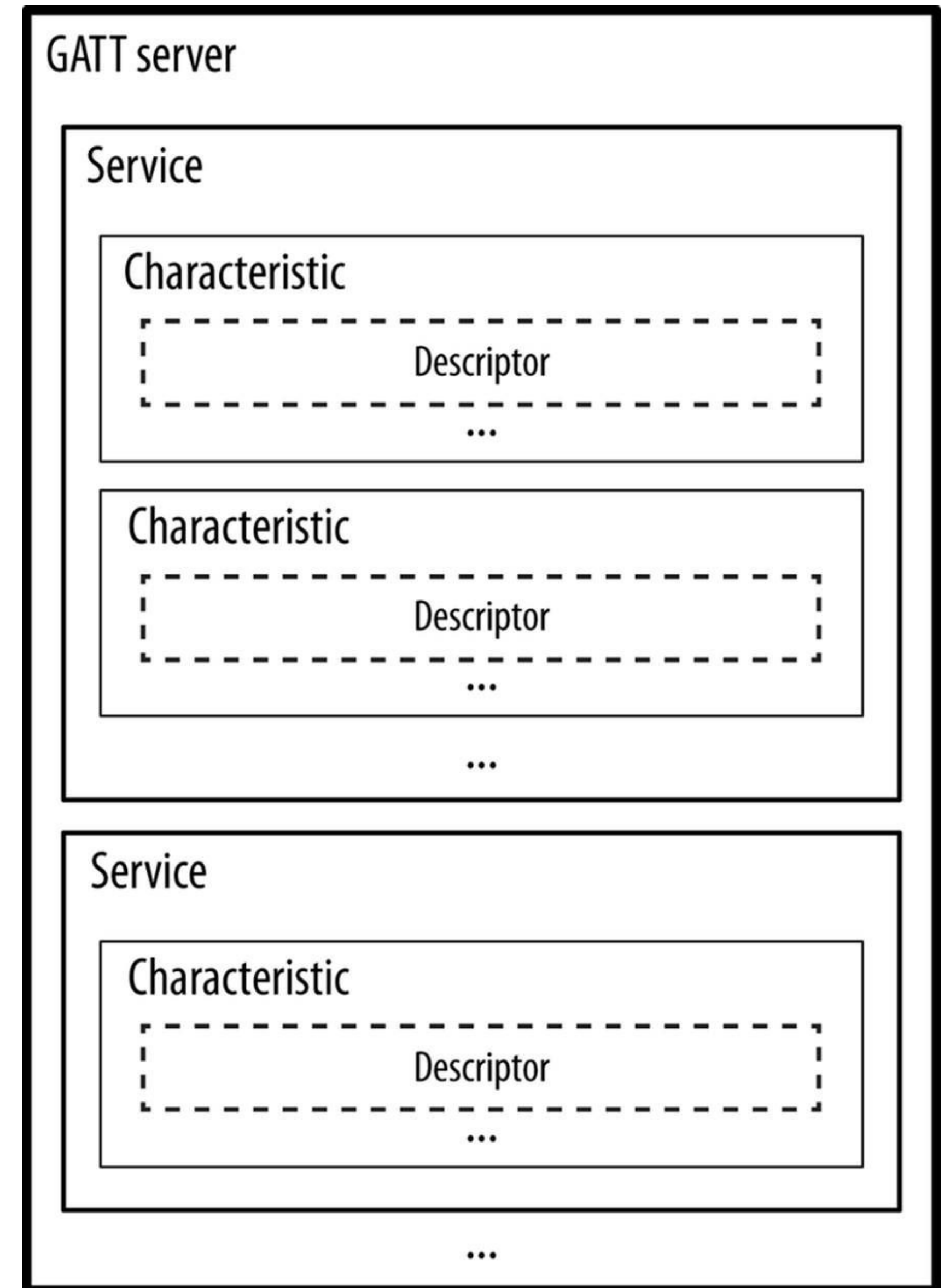
Using the Attribute Protocol, GATT groups services that encapsulate the behavior of part of a device and describes a use case, roles, and general behaviors based on the GATT functionality. Its service framework defines procedures and formats of services and their characteristics, including discovering, reading, writing, notifying, and indicating characteristics, as well as configuring the broadcast of characteristics. GATT is used only in Bluetooth LE implementations.



Source: <https://www.bluetooth.com/specifications/bluetooth-core-specification>

GATT

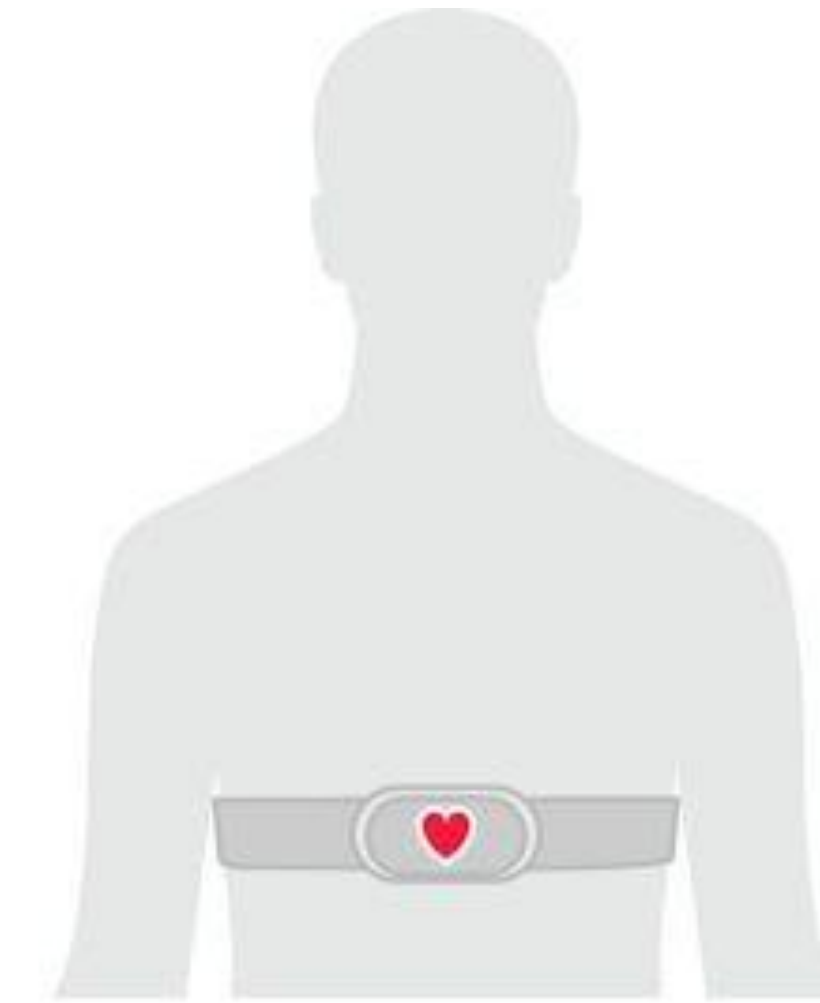
- Attributes in a GATT server are grouped into services.
- Each service can contain zero or more characteristics.
- Characteristics can include zero or more descriptors.



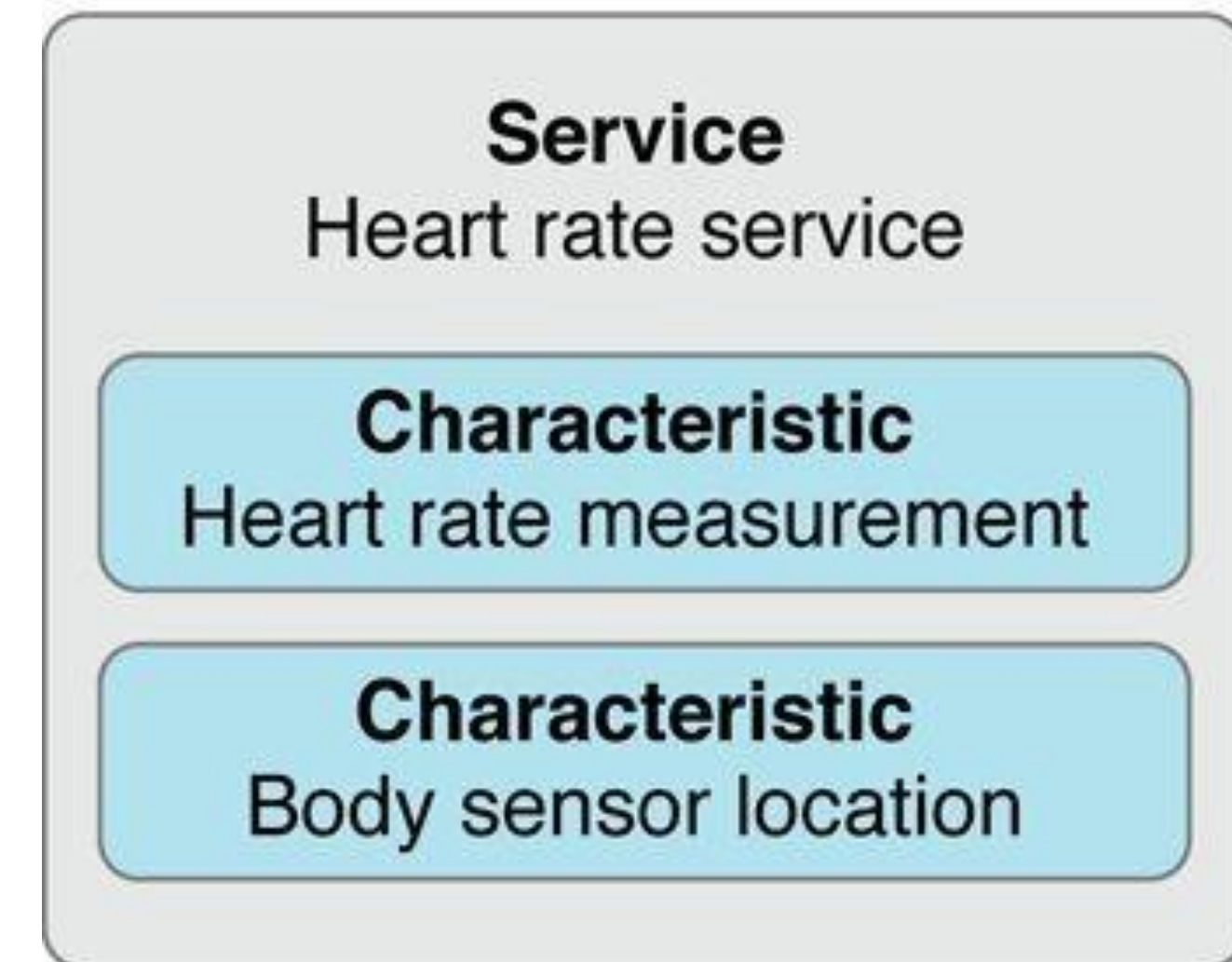
GATT

For example, a heart rate service may contain one characteristic that describes the intended body location of the device's heart rate sensor and another characteristic that transmits heart rate measurement data.

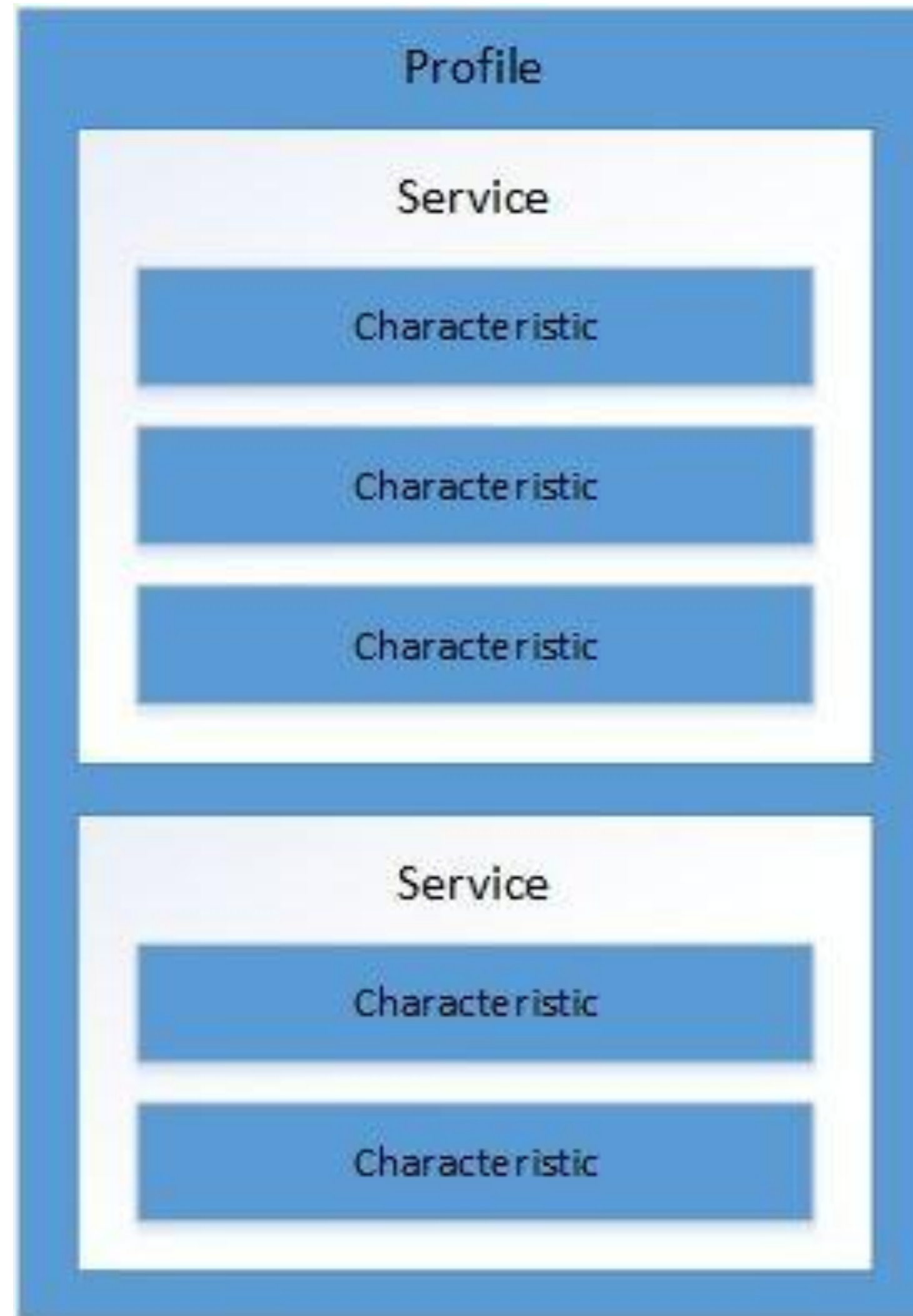
Source: <https://developer.apple.com>



Peripheral

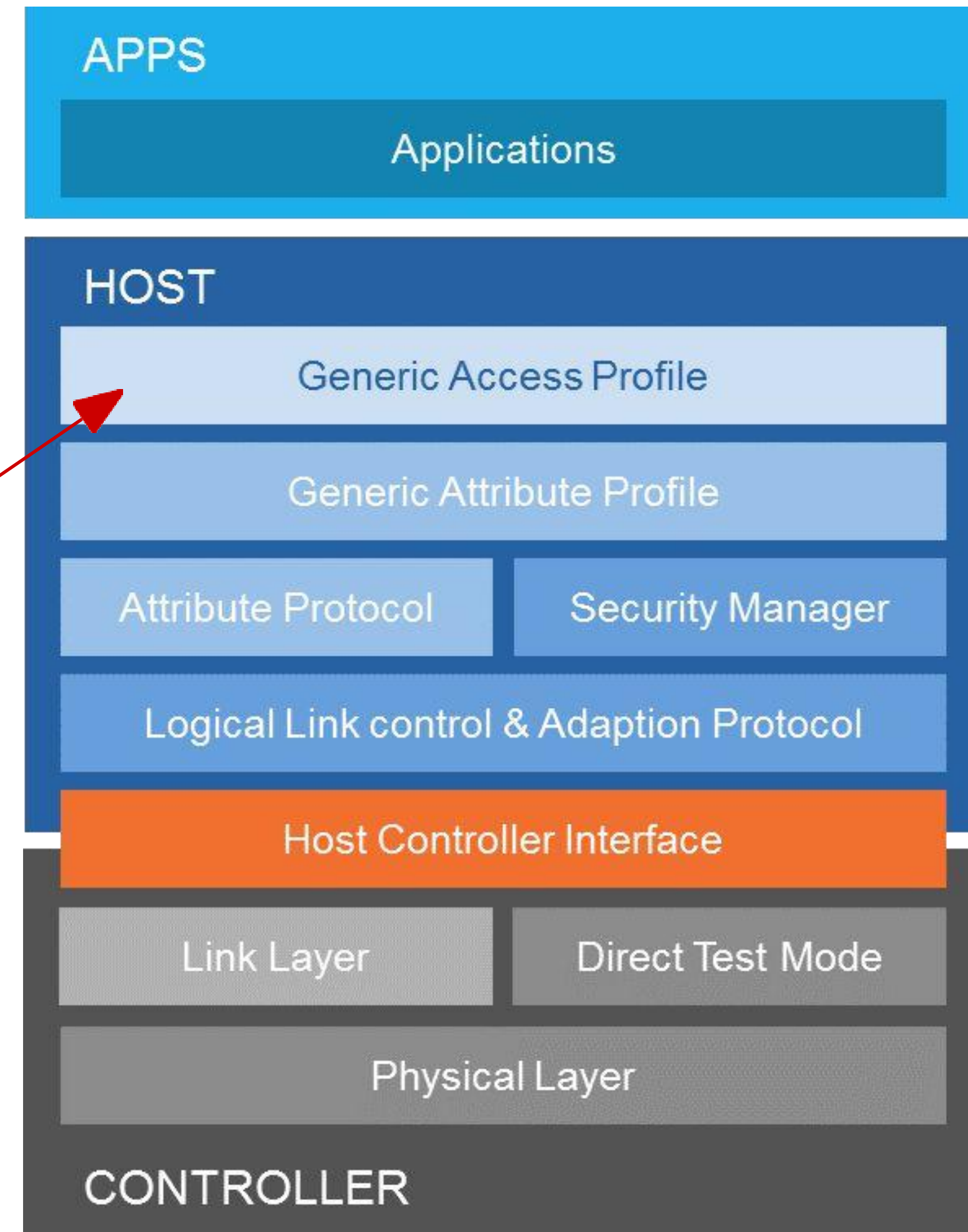


GATT: Profiles



BLE System Architecture

GAP works in conjunction with GATT in Bluetooth LE implementations to define the procedures and roles involved in the discovery of devices, sharing information, and managing link aspects of connecting to Bluetooth devices.

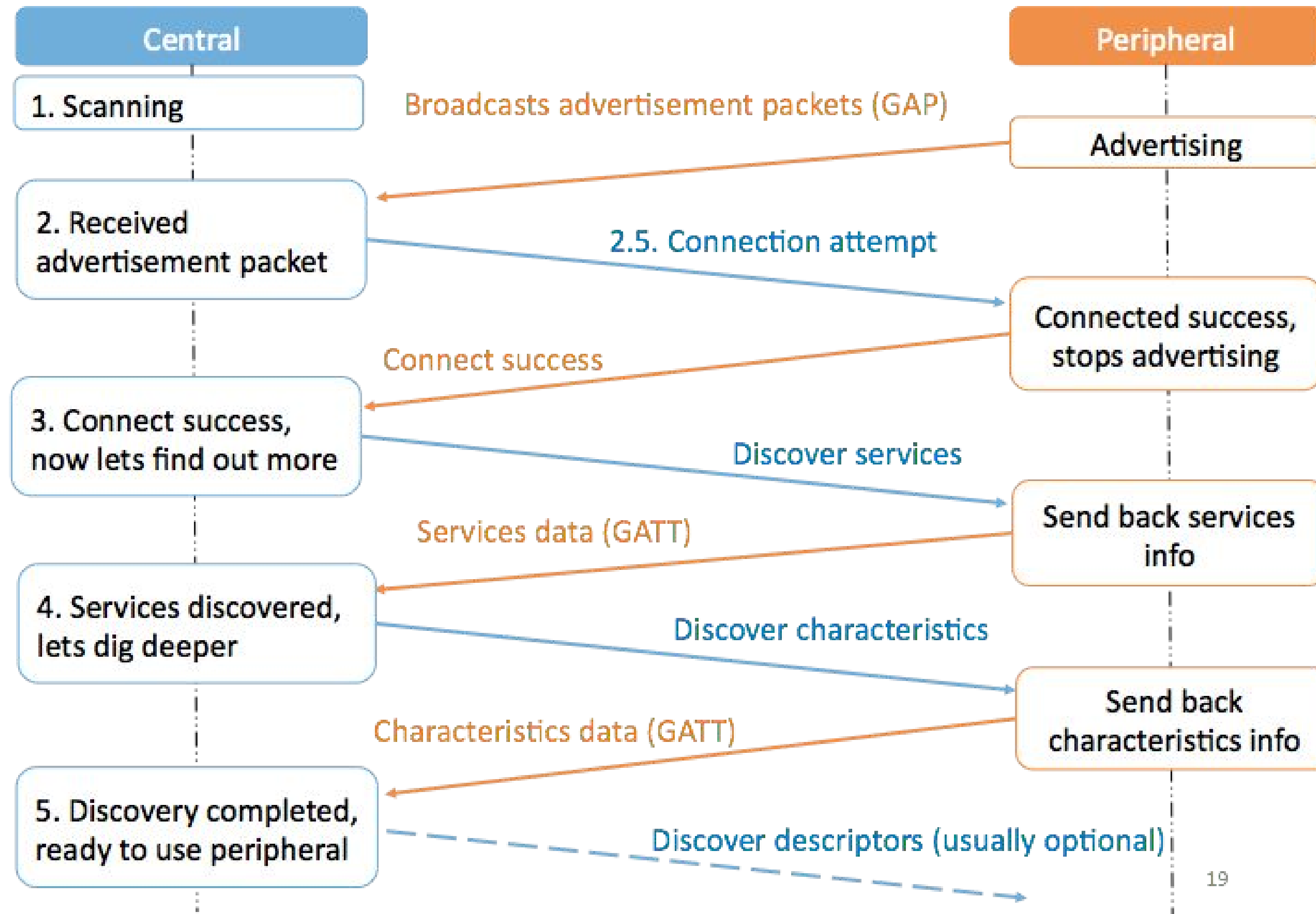


Source: <https://www.bluetooth.com/specifications/bluetooth-core-specification>

Generic Access Profile (GAP)

- Allows Bluetooth Low Energy devices to interoperate with each other.
- GAP specifies four roles that a device can adopt to join a BLE network:
 - **Broadcaster:** periodically sends out advertising packets with data.
 - **Observer:** listens for data embedded in advertising packets from broadcasting peers.
 - **Central:** corresponds to the Link Layer master. A device capable of establishing multiple connections to peers.
 - **Peripheral:** corresponds to the Link Layer slave. This role uses advertising packets to allow centrals to find it and, subsequently, to establish a connection with it.

GAP



Broadcaster use case

BLE beacons and the Physical Web:

<https://www.youtube.com/watch?v=1yaLPRgtIR0>

Eddystone format:

<https://developers.google.com/beacons/eddystone>

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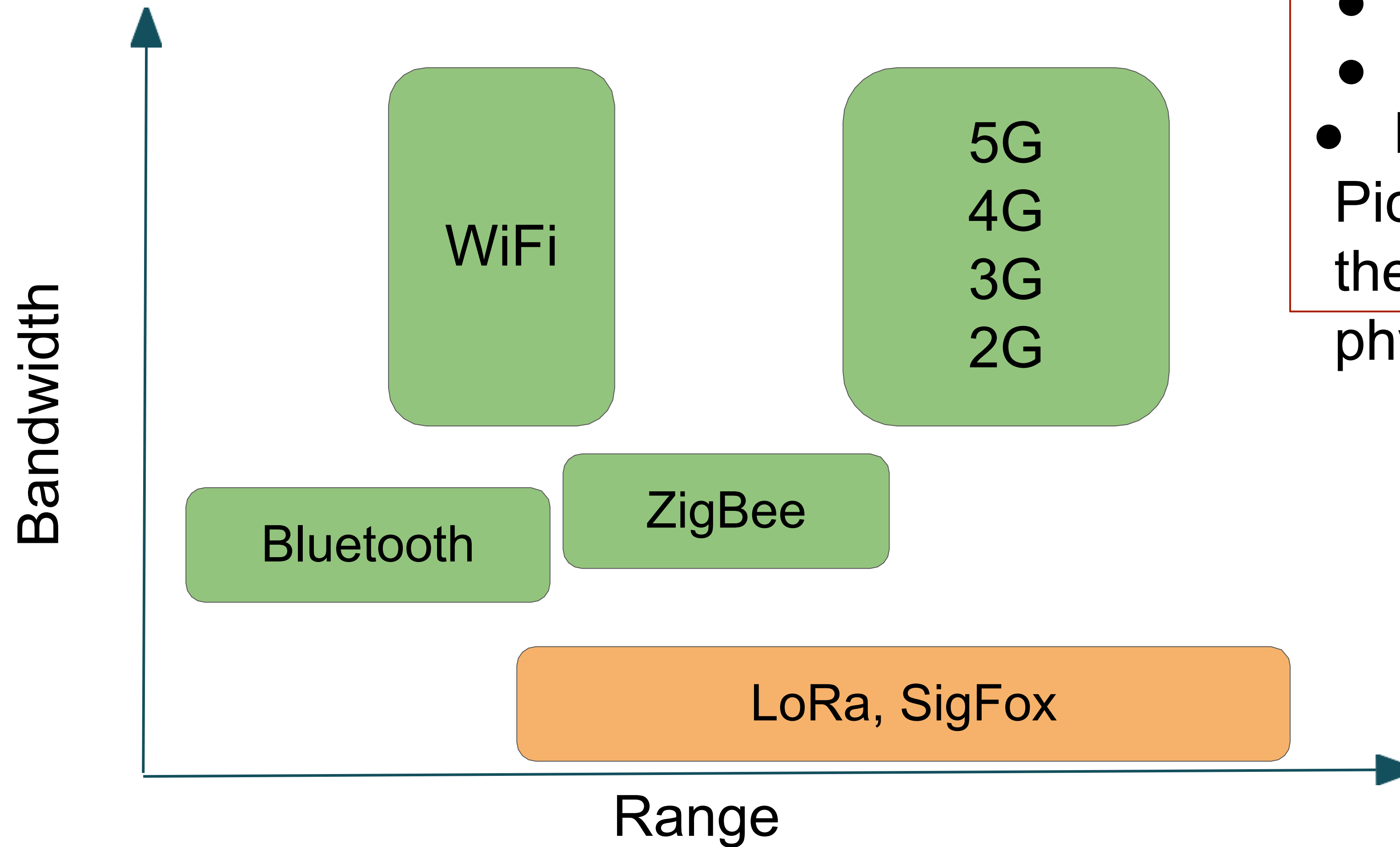
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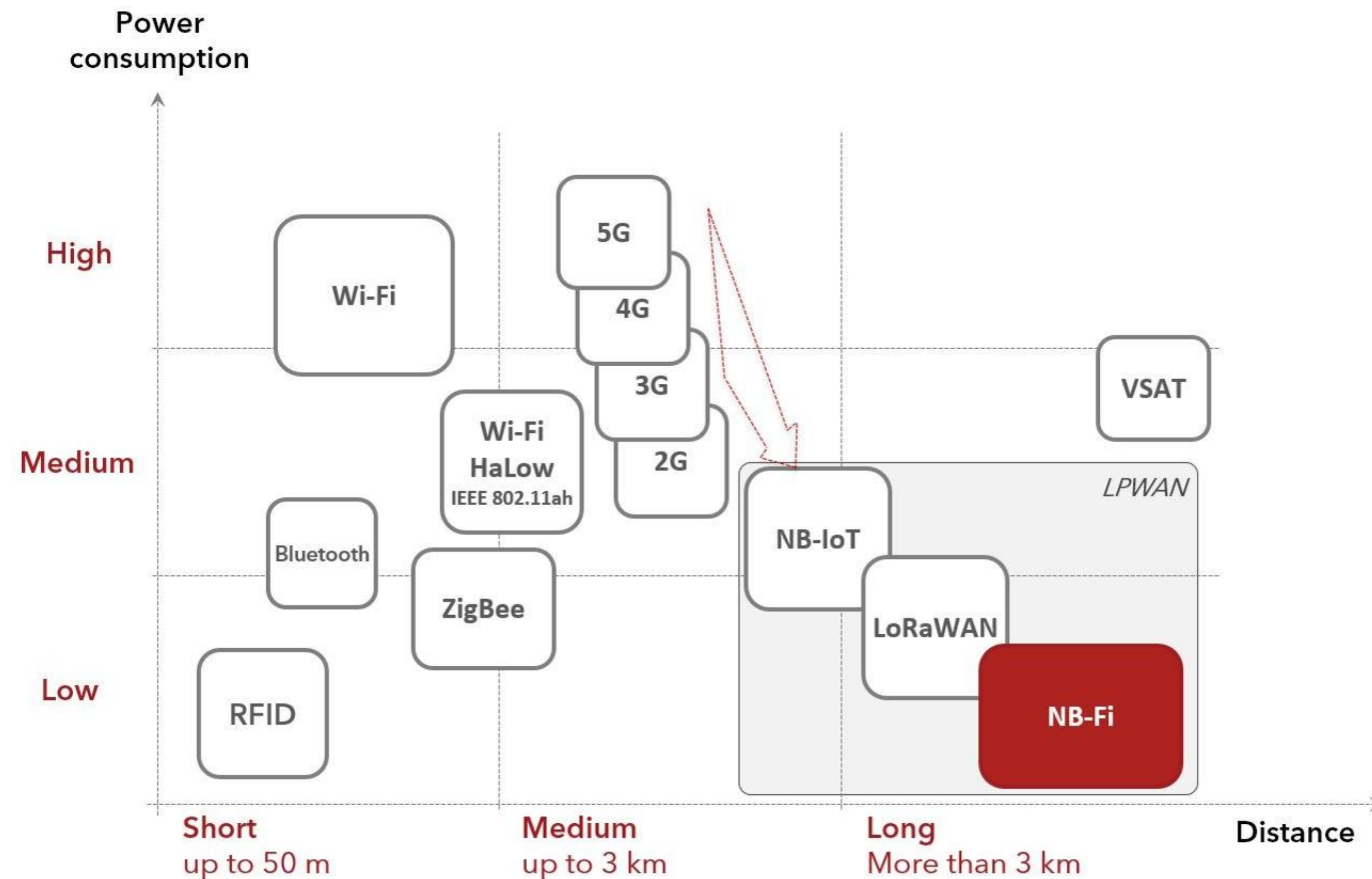
Low Power Wide Area Networks



- Long Distance
 - High Speed
 - Low Power
- Pick 2 (dictates the laws of physics)

Source: Thomas Telkamp, LoRa Crash Course

Low Power Wide Area Networks



- Long Distance
- High Speed
- Low Power Pick 2 (dictates the laws of physics)

Source: <https://waviot.com/technology/>

LoRa

- Wireless modulation technology
 - Chirp spread spectrum
- Physical layer for long range communications
- Low bandwidth
- Low battery usage (~ 10 years lifetime)
- Operates in the license-free ISM bands all around the world
 - 902 - 928 MHz in US. Power limit. No duty cycle limit.
 - 863 - 870 MHz in Europe. 1% duty cycle limit = 36 sec/hour in each sub-band for each end-device
 - 923 MHz in Asia
 - 915 - 928 MHz in Australia and Brazil

Source: Thomas Telkamp, LoRa Crash Course

LoRa

LoRaWAN® Formally Recognized as an ITU International Standard

15 December 2021 / by [Olivier Beaujard](#)



[Tweet](#)

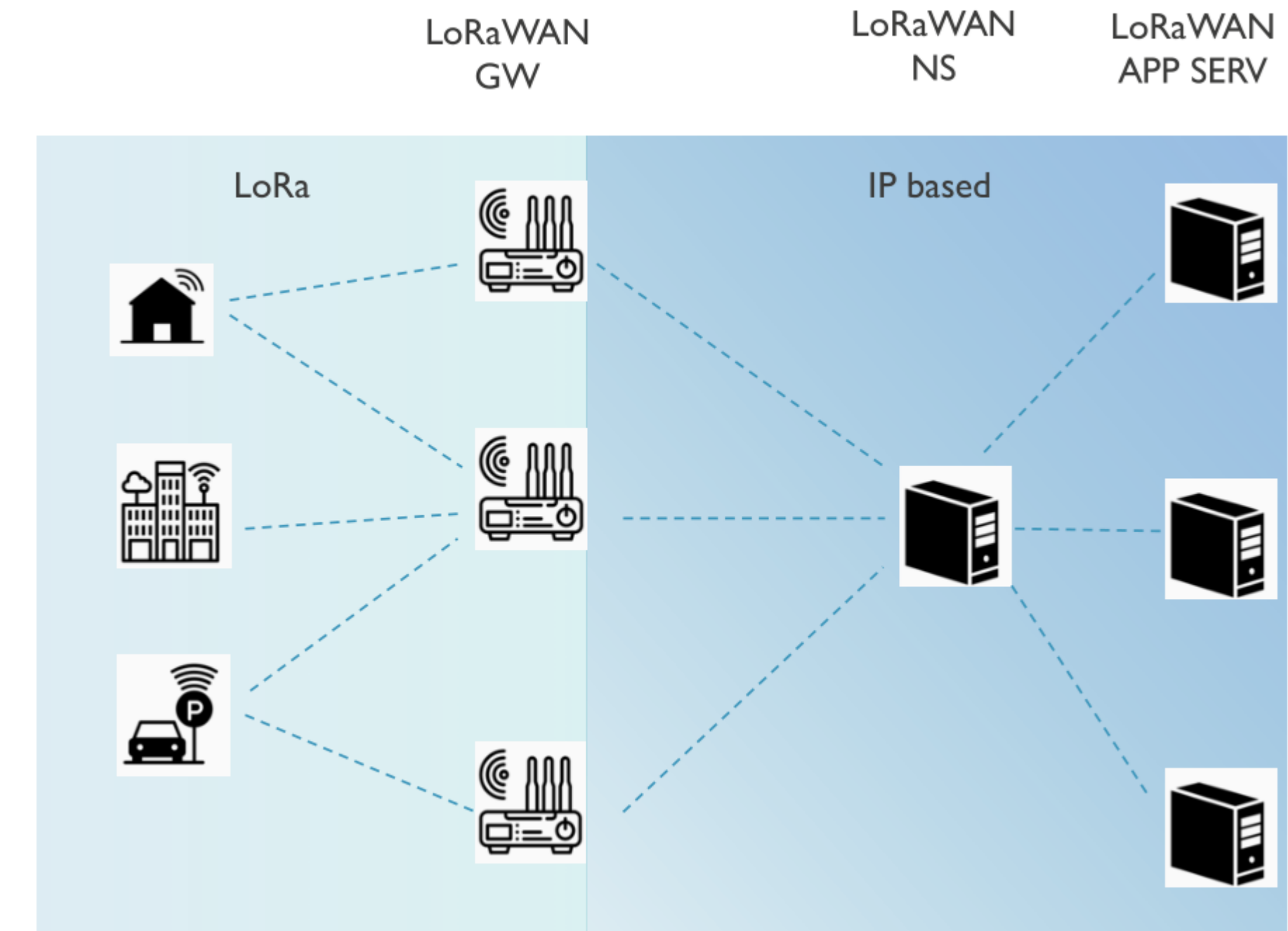
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The LoRaWAN® standard has been officially approved as a standard for low power wide area networking (LPWAN) by the [International Telecommunication Union](#) (ITU), the United Nations specialized agency for information and communication technologies.



Source: Haxhibeqiri, J., De Poorter, E., Moerman, I., & Hoebeke, J. (2018). A survey of LoRaWAN for IoT: From technology to application. *Sensors*, 18(11), 3995.

LoRa

LoRa Edge™ LR1110 Solution

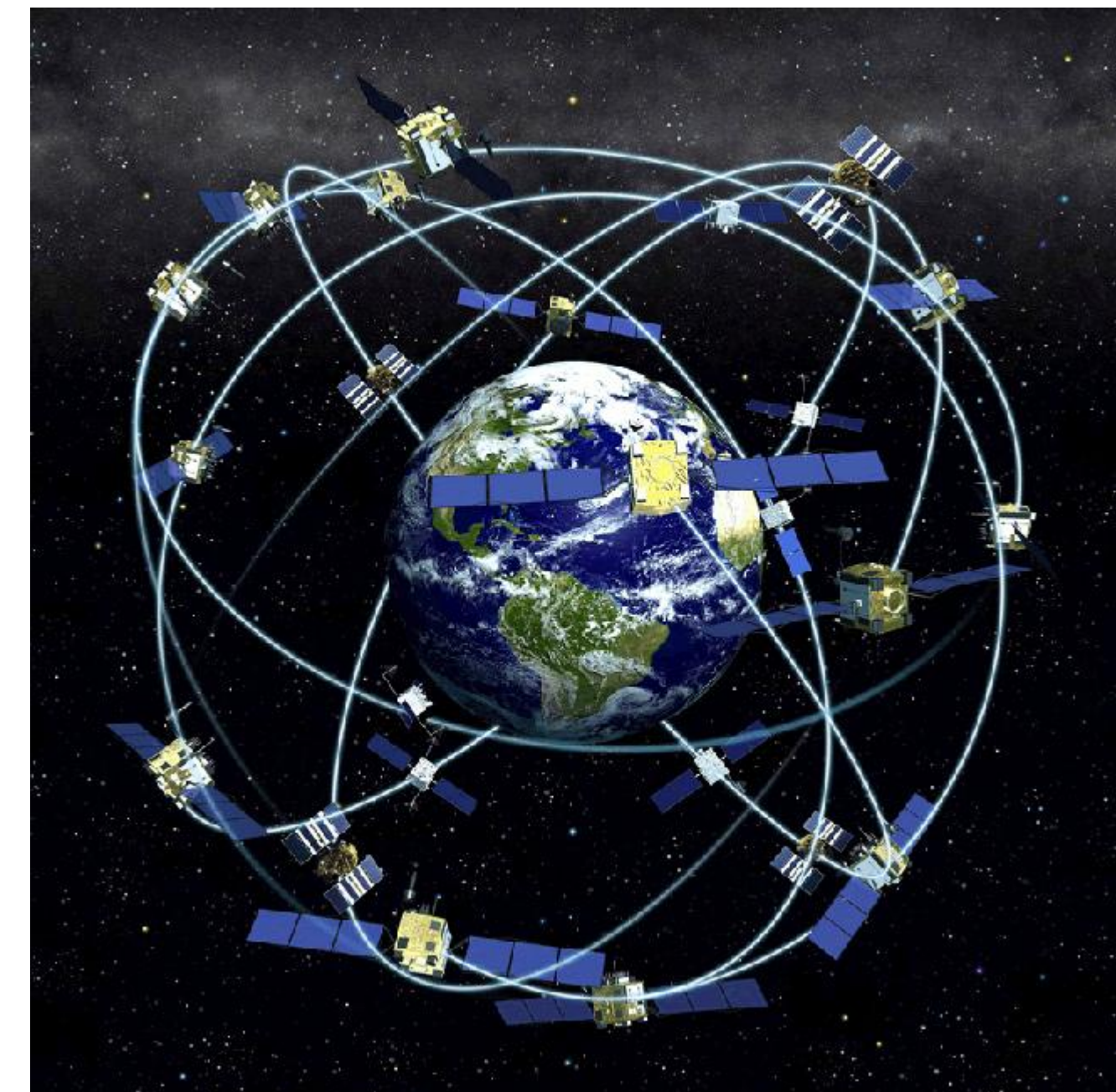


Power at one tenth of
competing solutions

Innovative Cloud-based solver

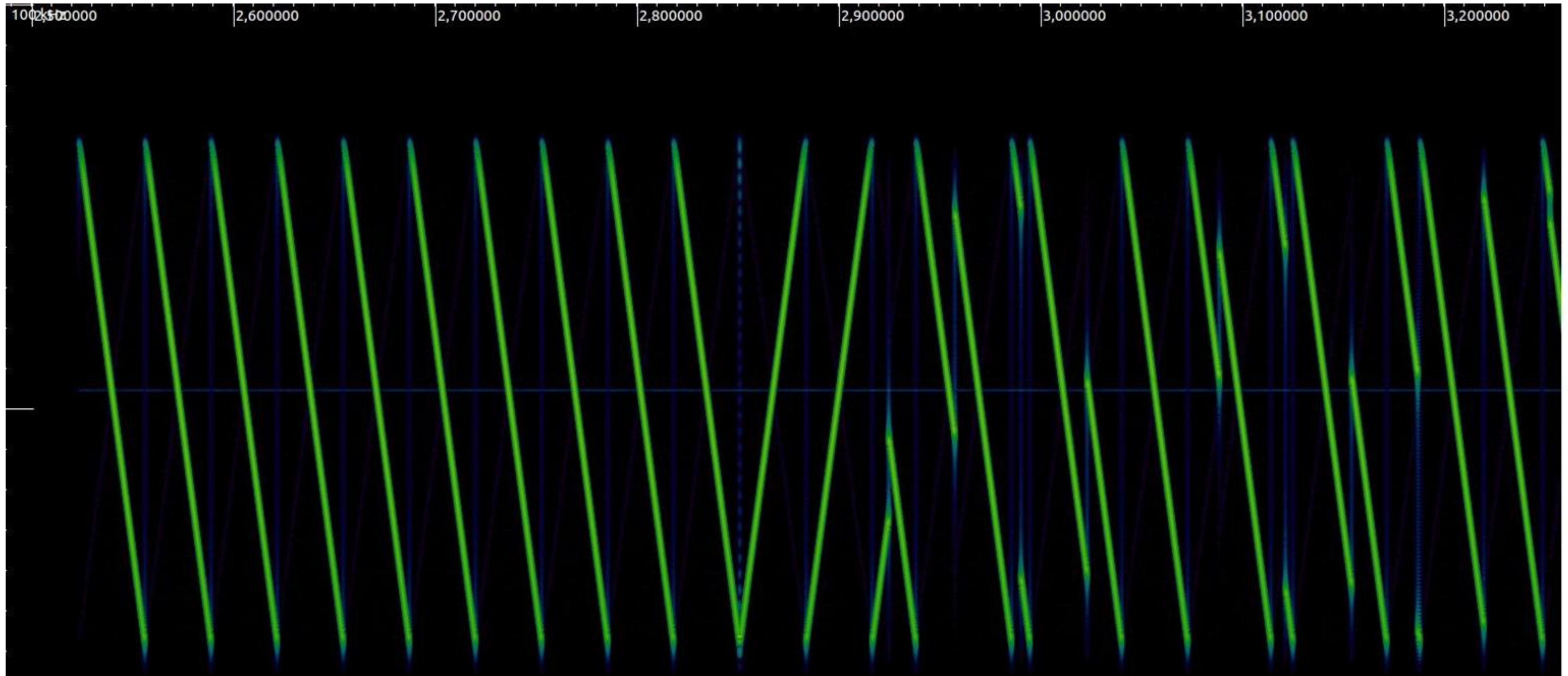
vs

GPS



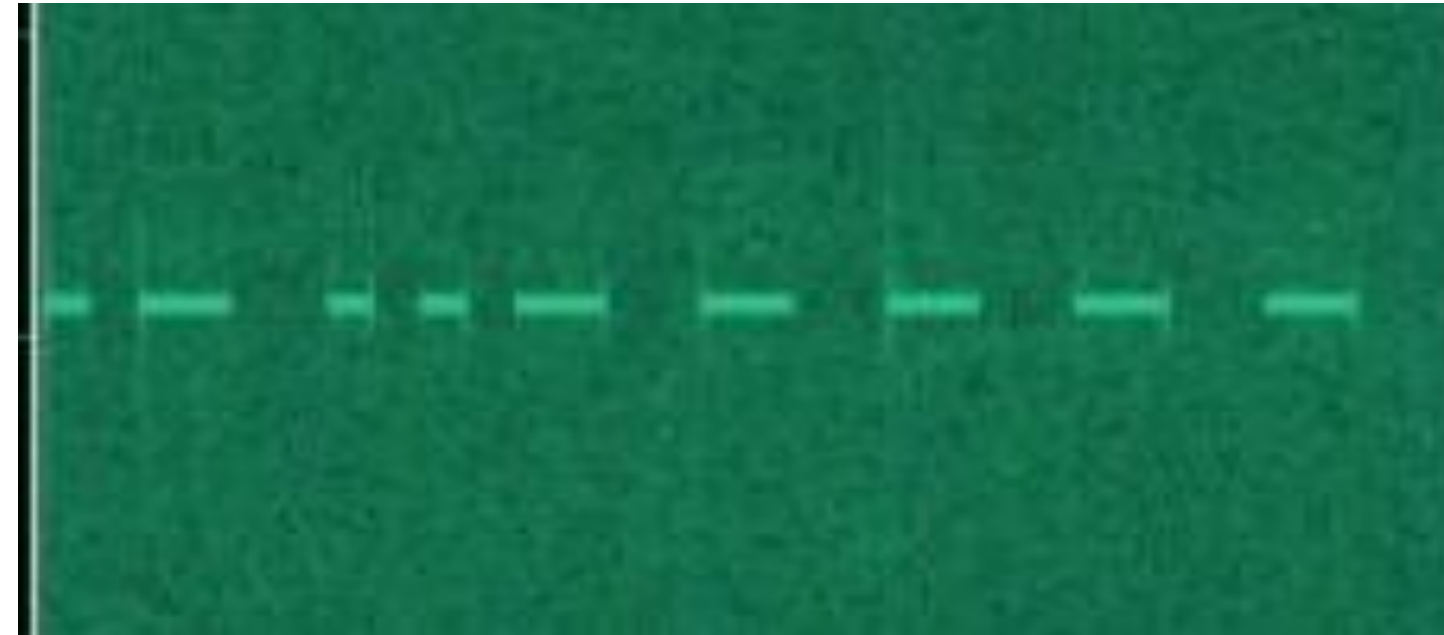
- Location

LoRa - Chirp spread spectrum



LoRa - Chirp spread spectrum

On-Off Keying

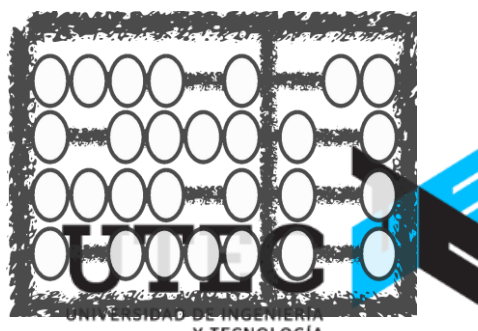


Frequency-Shift Keying

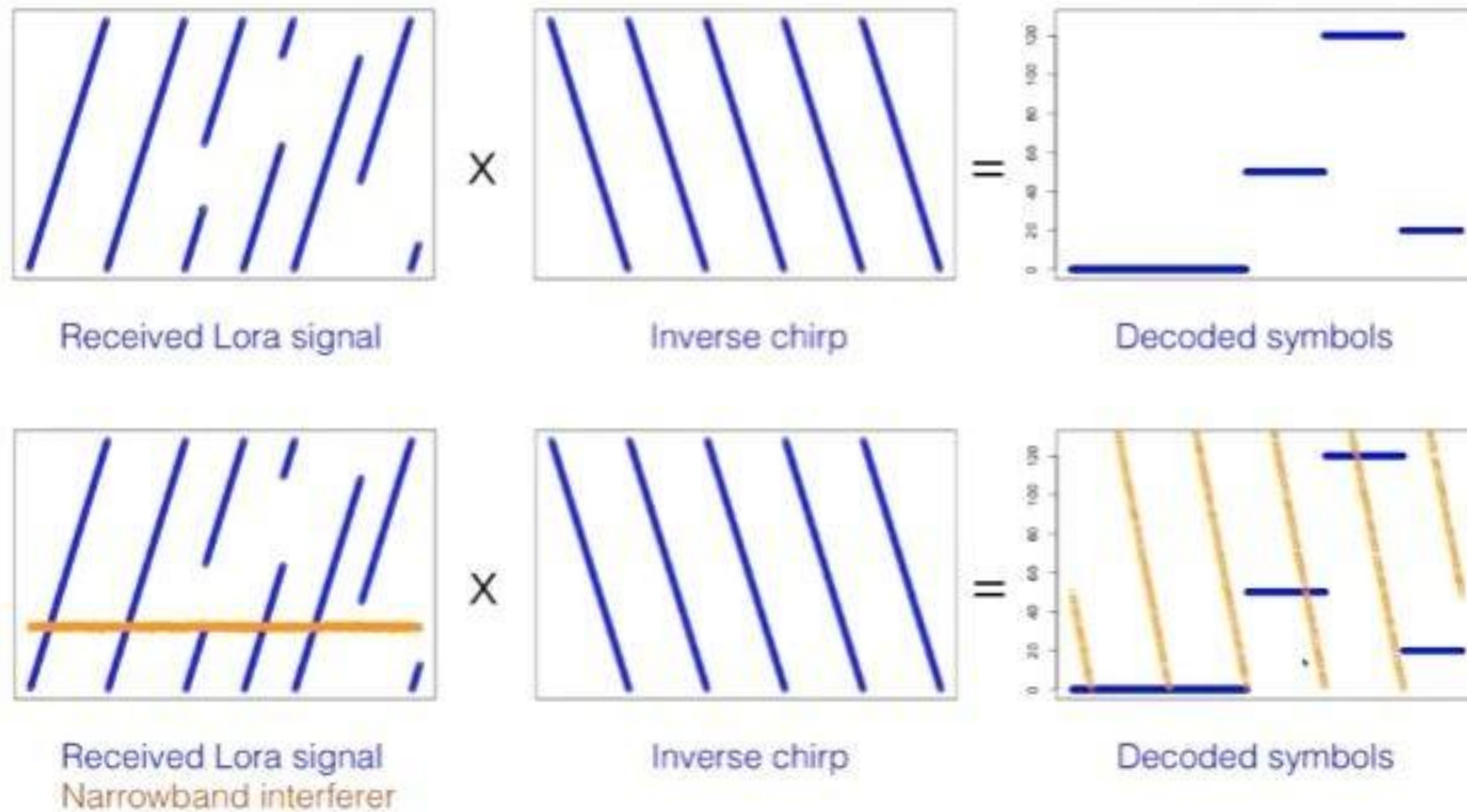


LoRa

Source: Thomas Telkamp, LoRa
Crash Course



LoRa - Chirp spread spectrum



Source: Thomas Telkamp, LoRa Crash Course

LoRaWAN

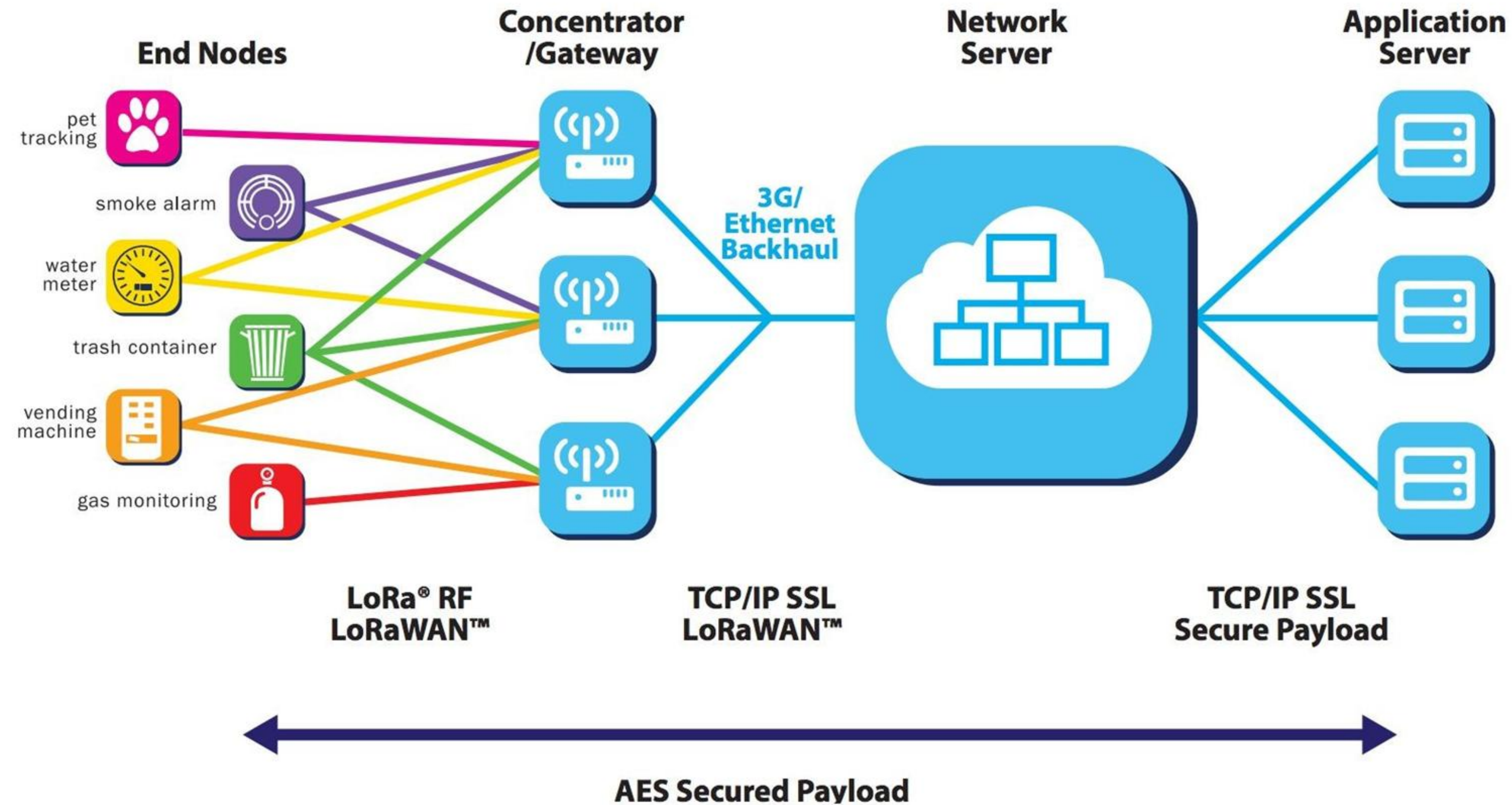
- Long Range Wide Area Network.
- Communications protocol and architecture that utilizes the LoRa physical layer
- Originally developed by Cyclos in France, acquired by Semtech corporation, which formed [LoRa Alliance](#).
 - Now 160+ members.
- V1.0 spec dated January 2015. Released to public July 2015.

Source: http://www.cse.wustl.edu/~jain/cse574-16/ftp/j_14ahl.pdf

LoRaWAN

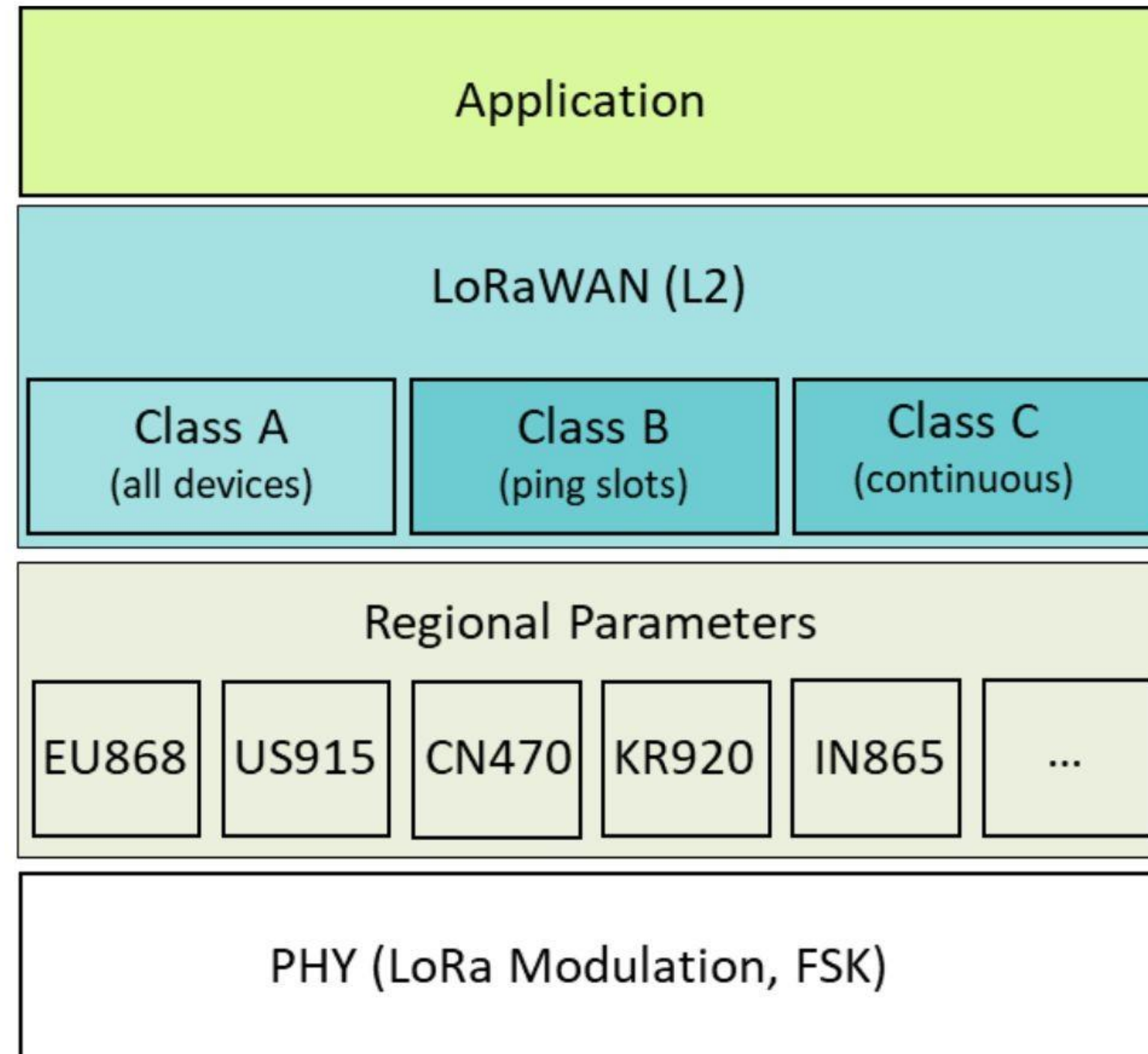
- Data rates are defined that range from 300 bps to 5.5kbps
- Supports
 - secure bidirectional communication,
 - mobility
 - localization

LoRaWAN: Topology



Source
LoRaW, ...

LoRaWAN: Protocol stack



Source: LoRaWAN Specification, v1.0.4.

Outline

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1

Introduction

BLE – All layers

Lora and the Physical Layer Comm

Conclusion

Conclusion

- We introduced two Physical and Link-layer communication protocols for modern IoT systems:
 - We describe BLE as an alternative to low power consumption comm-window-based communication protocol
 - We showed other low-power alternatives for communicating IoT nodes, Lora, LoraWAN and SigFox

Bibliography

- [LoRaWAN Specification](#), v1.0.4, 2020.
- A. Augustin, J. Yi, T. Clausen, W. M. Townsley. A study of LoRa: Long Range & Low Power Networks for the Internet of Things. Sensors 16(9), 2016.
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